

Review

Advancements and Prospects in Algal Biofuel Production: A Comprehensive Review

Halina Falfushynska ^{1,2} 

¹ Faculty of Economics, Anhalt University of Applied Sciences, Strenzfelder Allee 28, 06406 Bernburg, Germany; galyna.falfushynska@hs-anhalt.de

² ENERTRAG SE, Gut Dauerthal, 17291 Dauerthal, Germany

Abstract: Microalgae represent a valuable renewable resource for biofuel production due to their high lipid content, rapid growth rates, and non-competition with food resources. Both freshwater species like *Chlorella* and marine species such as *Dunaliella*, *Tetraselmis*, and *Nannochloropsis* are among the most commonly utilized candidates. This review provides a comprehensive overview of current cultivation and harvesting methodologies for microalgae in the context of biofuel production, emphasizing sustainable aviation fuel and biohydrogen. It synthesizes recent findings, technological advancements, and practical implementations to enhance the productive and economic viability of microalgae-based biofuels, highlighting their potential as a sustainable renewable energy source. Among the biofuels, sustainable aviation fuel and biohydrogen stand out as significant contributors to reducing greenhouse gas emissions. Technologies such as the oil-to-jet process and Fischer–Tropsch synthesis are being optimized to convert algal lipids into high-quality fuels. Biohydrogen offers several advantages, including the potential for negative CO₂ emissions and compatibility with existing hydrogen infrastructure. Despite the challenges associated with the high costs of cultivation and processing, advances in biotechnological methods and process engineering promise to overcome these barriers. This review highlights the importance of continued research and development to maximize the potential of microalgal biofuels in achieving sustainable energy goals and contributing to global efforts in mitigating climate change.

Keywords: algae; cultivation; harvesting; fatty acid composition; biodiesel; sustainable aviation fuel; biohydrogen



Citation: Falfushynska, H.

Advancements and Prospects in Algal Biofuel Production: A Comprehensive

Review. *Phycology* **2024**, *4*, 548–575.

[https://doi.org/10.3390/](https://doi.org/10.3390/phycology4040030)

[phycology4040030](https://doi.org/10.3390/phycology4040030)

Academic Editor: Marcin Dębowski

Received: 11 September 2024

Revised: 4 October 2024

Accepted: 9 October 2024

Published: 11 October 2024



Copyright: © 2024 by the author. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Algae represent a highly diverse group of photosynthetic organisms that can exist in both unicellular and multicellular forms. These organisms are categorized based on a range of morphological and ecological characteristics [1]. In aquatic environments, algae represent a group of primary producers, significantly contributing to oxygen production [2] and forming the basis of many aquatic food webs.

Beyond their ecological importance, algae are utilized in various health and wellness applications. Some would use algae for treating precancerous growths, boosting the immune system, improving the elemental composition of a body, increasing energy and metabolism, lowering cholesterol, preventing heart disease, healing wounds, and enhancing digestion and bowel health [3,4]. This wide array of health benefits has made algae a valuable component in both traditional and alternative medicinal practices. Moreover, algae hold significant promise in the field of sustainable energy [5]. Because the higher heating values of microalgae range from 14 to 24 MJ/kg, being comparable to those of lignocellulosic biomass [6], they serve as an extensive feedstock for biofuel production—a sector where they offer a cost-competitive alternative to conventional fossil fuels while being much more environmentally friendly [7–9]. The rapid growth rates and high lipid content of certain algae species enable the efficient harvesting of substantial biomass. This

makes algae a highly promising renewable resource not only for various biotechnological industries, ranging from biofuels to bioplastics and pharmaceuticals, but also for the energetic sector.

Energy consumption has significantly increased [10] due to urbanization, industrialization, and rising populations, pushing fossil fuels towards depletion. According to a modified Klass model, oil, coal, and gas reserves will last approximately 35, 107, and 37 years, respectively, suggesting coal may be the sole fossil fuel left by 2042 [11]. The energy sector, predominantly relying on fossil fuels, contributes over 75% of the EU's greenhouse gas emissions. Increasing renewable energy use is critical to reduce these emissions by at least 55% by 2030 and achieve climate neutrality by 2050 [12]. Germany aims to be more proactive and has announced an ambitious goal of reducing emissions by 80–95% by 2050 compared to 1990 levels [13].

Following its commitment to sustainable energy goals, research institutions and businesses have explored the implementation of various types of biomass for biofuel production. The utilization of biomass shows significant potential for achieving negative CO₂ emissions [14] compared to other renewable sources like wind and solar [15]. This initiative began with the production of ethanol from corn or sugarcane and has since progressed to include second-generation biofuels derived from non-food crops or agricultural residues. The field has further evolved into third-generation biofuels, primarily associated with algal biomass, and even into fourth-generation biofuels derived from genetically engineered plants and microorganisms [16]. These diverse feedstocks have been utilized to produce a wide array of biofuels, including biodiesel, bioethanol, biohydrogen, bio-oil, sustainable aviation fuel, and biogas, all with high efficiency. The oil conversion efficiency of different feedstocks follows the pattern microalgae (91%) > oil palm (3%) > coconut (1.5%) > avocado (1.4%) > jatropha (1.2%) > rapeseed/canola (1%). This clearly indicates that the oil yield from microalgae is much higher than that from other feedstocks [17]. Also, to produce 1 L of biodiesel, algae require approximately 300 to 1000 L of water, regardless of its quality. This demand is considerably lower compared to most first-generation feedstocks, which need around 5500 L for canola and 15,000 L for soybean [18].

Replacing all transportation fuel in the United States based on 2007 consumption would require 0.53 billion cubic meters of biodiesel annually, while Europe would need 0.4 billion cubic meters. Microalgae, with oil content ranging from 30% to 70%, need only 2.5% to 1.1% of total cropping area for complete fossil fuel replacement, making them a highly viable option for large-scale fuel production [19]. To meet this demand using microalgae with a productivity of 400,000 L per hectare, 9.25 million hectares of land would be necessary for Europe [20].

Despite the numerous benefits of biomass resources, especially for second-generation biofuels, their extensive application can lead to significant ecological issues such as deforestation and biodiversity loss. Conversely, algae offer a sustainable renewable energy alternative. However, challenges related to the cultivation, harvesting, extraction, and biofuel synthesis of algae need to be addressed, especially considering their energy- and resource-intensive nature. Addressing these challenges will be important for realizing the full potential of algae-based biofuels. Therefore, the goal of this review article is to provide a comprehensive overview of the current state of cultivation and harvesting methodologies utilized for microalgae as well as suitable approaches for the optimization of fatty acid composition and lipid productivity in the context of biofuel production with special emphases on sustainable aviation fuel and biohydrogen. This article also aimed to synthesize recent research findings, technological advancements, and practical implementations to enhance the productive and economic viability of microalgae-based biofuel as a sustainable renewable alternative.

2. Cultivation and Harvesting of Microalgae for Production of Algae Biofuel

2.1. Cultivation Systems for Microalgae-Based Biofuel Production

Microalgae have attracted significant attention as a sustainable resource for biodiesel production due to their high lipid content and rapid growth rate. The cultivation approach must be selected precisely to reach optimal results, with popular methods including photobioreactors, open pond systems, and hybrid systems [21]. The production of microalgae in Europe, which represents about 1% of the total algae production in the region, is primarily land-based and takes place in various systems: photobioreactors (71%), ponds (19%), and fermenters (10%). The common applications of microalgae include food supplements, nutraceuticals, cosmetics, wellbeing products, and animal feed [22]. Less than 22% of synthesized algal biomass in Europe is used for biofuel production [23].

Photobioreactors are closed systems that provide a controlled environment for the cultivation of microalgae. Photobioreactors offer several advantages, including better control over environmental parameters such as light, temperature, and CO₂ concentration [24]. The closed nature of photobioreactors reduces the risk of contamination and allows for the cultivation of high-density algae cultures. Different designs of photobioreactors, such as tubular, flat-panel, and columnar configurations, support various needs. Tubular and flat-panel photobioreactors are among the most commonly used for large-scale biodiesel production [17]. These configurations provide large surface areas for light exposure. A comparative study showed that in a 10-day cultivation experiment, the peak CO₂ fixation rate for a tangent double-tube photobioreactor was 0.8 g·L⁻¹·d⁻¹, increasing by 33.3% compared to a concentric double-tube photobioreactor and by 293% compared to a traditional tube photobioreactor. Furthermore, the microalgal biomass yield increased by 51.2% and 124.8%, respectively [25]. Despite some benefits, the primary challenge with photobioreactors is their high capital and operational costs, which can be barriers to scalability [26].

Open pond systems, including raceway ponds, reflect another widely used approach for microalgae cultivation [21]. These systems consist of shallow, artificial pools where algae are cultivated in water bodies mixed typically by paddle wheels or other mechanical means. Open pond systems are cost-effective and simpler to construct compared to photobioreactors. However, they are more susceptible to contamination by other microorganisms and environmental conditions such as evaporation, temperature fluctuations, and pollutant ingress. While the lipid yield in open pond systems is generally lower than in photobioreactors, they remain popular due to their economic feasibility. A recent example highlights the potential of optimized open pond systems. In particular, about 19.98 wt % lipid was extracted from *Chlorella minutissima* cultured in raceway open ponds using the Soxhlet extraction method. The biodiesel produced met ASTM D6751 [27] and EN 14214 standard [28] specifications, except for oxidation stability which, after the addition of 1000 ppm propyl gallate, improved from 3.0 h to 10.5 h [29].

Open ponds, raceway ponds, and tanks are efficient when used for wastewater treatment. Many microalgae species that grow in these open systems can achieve low energy costs, making them an attractive option for sustainable wastewater management. As an example, the most environmentally favorable scenario involved using *Desmodesmus subspicatus* in an open raceway pond to remediate NPK-rich wastewater. This process was followed by centrifugation and drying, resulting in clean water and the production of biomass, which can be further utilized, specifically for diesel production [30]. The efficiency of such ponds becomes particularly optimal when considering the design of the pond, the hydrodynamics of the algal culture, and the growth conditions [31].

To maximize the advantages and mitigate the limitations of both photobioreactors and open pond systems, hybrid systems have been explored. These systems involve the integration of closed photobioreactors with open ponds, enabling initial growth phases in controlled environments before transferring cultures to open systems for large-scale biomass production. This approach takes advantage of the high productivity and controlled conditions of photobioreactors while using the cost-effectiveness and simplicity of open

pond systems, potentially leading to enhanced lipid accumulation and a scalable and economically viable solution for biodiesel production from microalgae.

A study on a putative biodiesel production plant utilizing the freshwater alga *Chlorella vulgaris* demonstrated the substantial environmental benefits of such hybrid systems [32]. The Life Cycle Assessment of this configuration revealed that the hybrid method could achieve significant reductions in global warming potential and fossil-energy requirements compared to traditional fossil-derived diesel. Specifically, there were 42% and 38% savings in global warming potential and fossil-energy requirements, respectively, for the hybrid system when producing one ton of biodiesel. Moreover, under mixotrophic growth conditions, the savings reached up to 76% for global warming potential and 75% for fossil-energy requirements, underscoring the ecological advantages of this approach [32].

The success of outdoor algae cultivation systems is often challenged by the need to maintain desirable species while mitigating the adverse effects of invasive algal species. A hybrid system can effectively address this challenge. In a 47-day operational study, the hybrid setup enabled the continuous provision of a desirable algal inoculum, maintaining the predominant growth of the target microalga *Parachlorella* sp. [33]. The enhanced performance of the hybrid system was evident, with the small-scale open raceway ponds showing a 40% increase in algal biomass and a 62% rise in lipid productivity compared to the conventional approach. The addition of CO₂ further amplified the productivity, aligning with previous studies that advocate for CO₂ management as an effective strategy for increasing industrial algal cultivation yields [34].

Another viable option within hybrid systems involves the integration of biofilm reactors, which can alleviate harvesting bottlenecks and improve biomass production. By coupling biofilm reactors with high-rate ponds during wastewater treatment, a higher biomass recovery efficiency of up to 40% can be achieved [35].

2.2. Cultivation Modes for Enhanced Microalgae Biomass and Biodiesel Production

The cultivation of microalgae using heterotrophic, mixotrophic, and photoheterotrophic modes represents promising approaches for enhancing biomass productivity and reducing the costs associated with phototrophic cultivation [36]. Despite the current reliance on phototrophic cultivation and fermenters using glucose, these methods often face economic and technological challenges that limit their broad application.

Heterotrophic cultivation involves growing microalgae in the dark with organic carbon sources, leading to higher biomass productivity as compared to phototrophic modes [36]. This approach bypasses the low efficiency of photosynthesis and mitigates the challenges of providing and managing CO₂. Mixotrophic cultivation, on the other hand, incorporates both organic carbon sources and light, allowing microalgae to utilize both photosynthesis and respiration for growth [37]. Photoheterotrophic cultivation also relies on organic carbon sources along with light but leverages specific metabolic pathways that cannot fully function under strictly mixotrophic or heterotrophic conditions.

A promising substrate for the heterotrophic and mixotrophic cultivation of microalgae is acetate, which can potentially reduce costs and environmental impacts. Acetate is easily accessible and can be synthesized through various renewable processes, creating a bridge between industrial facilities and bio-based systems. Proietti Tocca et al. (2024) [38] highlight alternative processes for acetate synthesis, such as aerobic and anaerobic fermentations, thermochemical treatments, and microbial electrosynthesis. These processes offer distinct advantages and challenges that can influence the practicality and sustainability of acetate as a microalgae substrate [38].

Heterotrophic cultivation modes exhibit significant improvements in biomass yields due to the highly efficient utilization of organic carbon substrates. For instance, Grubišić et al. (2024) [39] highlighted that under heterotrophic growth conditions, biomass productivity (2.385 g L⁻¹ day⁻¹) and lipid productivity (0.339 g L⁻¹ day⁻¹) of *Chlorella vulgaris* S2 increased by approximately 50 and 60 times, respectively, compared to fed-batch phototrophic cultivation. Moreover, mixotrophic cultivation with a controlled carbon-to-

nitrogen (C:N) ratio could enhance desirable biomass and/or lipid production. Specifically, a C:N ratio of 30 mol mol⁻¹ was found to favor lipid synthesis, whereas a C:N ratio of 10 mol mol⁻¹ promoted carbohydrate synthesis. The highest lipid and biomass productivities (2.238 and 0.458 g L⁻¹ day⁻¹, respectively) were achieved under mixotrophic conditions with a C:N ratio of 50 mol mol⁻¹ and a glucose concentration of 50 g L⁻¹ [39]. Similarly, achieving the proper balance of carbon, nitrogen, and vitamins in the media can induce lipid accumulation of up to 77% of cell biomass in *Arthrospira platensis*, *Nannochloropsis* sp., and *Spirulina* sp. [40]. Additionally, acetate enhances biomass productivity under both mixotrophic and heterotrophic conditions when it is used as a carbon source by integrating seamlessly into metabolic pathways.

Moreover, mixotrophic cultivation offers significant advantages in adapting to various microelement-enriched substrates. This is exemplified by the work of Youssef et al. (2024) [41], who investigated the use of cheese whey as an organic carbon source for cultivating alkaliphilic microalgae. Their research demonstrated significant enhancements in both biomass productivity and lipid content, thereby supporting biofuel production while using low-cost substrates [41]. Specifically, *T. obliquus* grown under conditions of 3.5% (v/v) whey, pH 10.0, and 0.5 g L⁻¹ NaNO₃ achieved biomass production of 48.69 mg L⁻¹ day⁻¹ and lipid production of 20.64 mg L⁻¹ day⁻¹. In comparison, *Cyanotheca* cultivated under similar conditions—4.5% (v/v) whey, pH 9.0, and 1.0 g L⁻¹ NaNO₃—produced a comparable biomass yield of 52.78 mg L⁻¹ day⁻¹ but exhibited a lipid content that was half as much, at 11.42 mg L⁻¹ day⁻¹ [41]. Notably, the lipids produced under these conditions were rich in saturated fatty acids and monounsaturated fatty acids, with no polyunsaturated fatty acids present in either microalgae. These findings highlight the importance of selecting appropriate initial conditions and algae strains to maximize lipid and FAME production for efficient biofuel generation.

Despite the clear advantages of heterotrophic and mixotrophic modes, challenges related to large-scale implementation remain. Abreu et al. (2022) emphasize the need for further research to explore the potentials fully and address the constraints related to organic carbon substrate supply, metabolic engineering, and process design [36]. Sustainable and economically viable cultivation systems for microalgae require a comprehensive understanding of the underlying biology and engineering principles involved.

2.3. Wastewater-Based Microalgae Cultivation: Achieving Dual Benefits

In recent years, the cultivation of microalgae using wastewater has been pointed to as a promising approach for sustainable biofuel production and environmental management. Microalgae can grow heterotrophically or mixotrophically, enabling them to grow in nutrient-rich wastewaters and efficiently remove nutrients while producing biofuels. Heterotrophic microalgae metabolize organic carbon sources within the wastewater, leading to the removal of pollutants such as nitrogen and phosphorus. This process not only purifies wastewater but also produces biomass that can be converted into biodiesel, biogas, or other bio-products [42]. In particular, anaerobic digestates obtained from agro-waste or sewage sludge treatment result in an enhancement in lipid production per volume in *Chlorella vulgaris* cultures grown. On the other hand, mixotrophic microalgae can photosynthesize, capturing CO₂ from the atmosphere while simultaneously treating wastewater, thus contributing to carbon sequestration and nutrient removal with high efficiency (60–100%) [43].

A diverse range of wastewaters have been explored for microalgae cultivation, including municipal, industrial, aquaculture, and pharmaceutical wastewaters. Each type of wastewater provides a unique nutrient profile that can support the growth of different microalgal species. For example, *Chlorella vulgaris* grown in a closed photobioreactor on anaerobic digestates from agro-waste or sewage sludge can increase lipid production by up to 300%, while microalgae cultivated in swine or poultry wastewater exhibit enhanced biomass accumulation [42,44]. It is worth noting that high nutrient wastewater content elevates final biomass concentrations but does not necessarily enhance biomass productivity [44].

One study highlighted the potential of integrating microalgae cultivation with waste streams from a sugarcane-processing factory in Ethiopia. This integration led to the production of biodiesel, upgraded biogas, and bio-fertilizer, demonstrating a positive energy balance and substantial environmental benefits. The process yielded 188 tons of biodiesel, 1,974,882 m³ of upgraded biogas, and 42 tons of bio-fertilizer per year, and vinasse from the ethanol factory served as the primary nutrient source for microalgae cultivation [45]. Similarly, *Oocystis pusilla* showed the highest growth in a 100% wastewater medium, exhibiting increased biomass and lipid content compared to the KC medium. Conversely, *Chlorococcus infusionum* was unable to survive in these conditions. The highest biomass and lipid productivities were achieved at a TDS concentration of 3000 ppm, leading to a 28% increase in biomass (2.50 g/L) and a remarkable 158% increase in lipid yield (536.88 mg/g) compared to the KC medium [46].

However, the scalability of wastewater-based microalgae cultivation poses significant challenges. High costs associated with system construction, CO₂ gas transportation, and biomass harvesting, coupled with varied nutrient concentrations and retention times, affect productivity [44]. For instance, although high-rate algal ponds can remove up to 90% of NH₄, 70% of COD, and 50% of PO₄ content, maintaining optimal conditions for large-scale operations remains difficult. Moreover, pilot-scale studies have shown that microbial-specific growth rates (~0.54 day⁻¹) are halved compared to laboratory experiments, underscoring the need for effective scale-up strategies [44]. This issue can partly be mitigated by constructing robust microalgae–bacteria consortia, maximizing solar illumination, and ensuring a balanced provision of limiting nutrients, such as NH₄ and CO₂.

Therefore, microalgae present a sustainable and promising resource for biodiesel production, with various cultivation methods offering distinct advantages. While photobioreactors and open pond systems have their respective benefits and challenges, hybrid systems and non-phototrophic cultivation modes provide avenues for enhanced productivity and cost-efficiency. Wastewater-based microalgae cultivation highlights a dual benefit of biofuel production and environmental management. Integrating these systems with industrial and agricultural practices could pave the way for sustainable resource utilization and environmental protection, marking a significant step toward a green energy future.

2.4. Harvesting

Harvesting imposes a major constraint on microalgal downstream processes and the cost-effective production of various high-value products [47]. The primary aim of the harvesting method is to remove a maximum quantity of growth medium from the microalgal biomass, thereby facilitating an efficient extraction process [48]. This step is crucial as it directly impacts the economic viability and overall efficiency of biofuel production.

Various harvesting–dewatering techniques have been thoroughly investigated and validated, including centrifugation, coagulation, flocculation, flotation, sedimentation, screening, and filtration [49]. Each of these methods offers unique advantages and challenges in terms of efficiency, cost, and scalability.

Centrifugation is a proven harvesting–dewatering technology utilized across several industries due to its capability to achieve a high solid concentration, often reaching up to 10–20%. However, while efficient, centrifugation is an energy-intensive process. The high operational cost associated with centrifugation significantly affects the production economics of various microalgal products. Specifically, the energy input involved in centrifugation can make current commercial microalgal biodiesel production economically unfeasible, sometimes accounting for up to 50% of the total biofuel production cost [49]. Researchers are exploring the combination of centrifugation with flocculation or other cost-effective pre-concentration methods to mitigate energy consumption significantly [48]. Additionally, utilizing solar energy for biomass drying is being considered as an alternative. However, this approach's main drawback lies in its requirement for sufficient time and space for optimal and efficient processing [50].

Coagulation and flocculation are preferred techniques for large-scale harvesting of microalgae, primarily when used in biomass applications and biofuel production. Flocculation is typically used in conjunction with other harvesting methods to enhance efficiency [51]. By aggregating algal cells into larger particles, flocculation increases the rate of settling or flotation [52]. Coagulation, on the other hand, involves destabilizing a colloidal suspension using coagulants. Various chemical coagulants such as aluminum sulfate, polyaluminum chloride, ferric chloride, and ferric sulfate [53] are effective, with biological coagulants like filamentous fungus *A. fumigatus* also showing promise [48]. Laboratory studies demonstrate high flocculation efficiencies for microalgal species; for instance, *Scenedesmus obliquus* showed an efficiency of 83.2% at pH 12 after 1 h, while FeCl₃, alum, and chitosan resulted in efficiencies of 80.2%, 95%, and 91% respectively, at specific concentrations after 1 h [47]. Similar results were observed for *Chlorella vulgaris*, supporting the effectiveness of optimal doses of chitosan and aluminum sulfate in achieving more than 90% biomass recovery with a sedimentation time of 10 min [54].

Flotation is another promising method, often faster than sedimentation for various microalgal species. Some microalgae naturally float, but flotation efficiency can be enhanced using air bubbles [55]. Both conventional and non-conventional flotation technologies, such as electroflotation and dissolved air flotation, are emerging as promising approaches. Dissolved air flotation, extensively used for sludge thickening in wastewater treatment plants, is also applicable to microalgal biomass due to its ability to exploit microalgae's natural buoyancy and the formation of low-density flocs through coagulation. This method therefore harnesses the algal cells' natural self-floating tendency for efficient biomass recovery [53].

3. Lipid Productivity and Fatty Acid Composition

3.1. Fatty Acid Composition of Microalgae as the Important Property of Biodiesel Feedstock

In microalgal cells, lipids exist in the form of structural and storage lipids among them droplets. These lipids are classified into two categories: polar and non-polar. Non-polar lipids, also known as neutral or storage lipids, include triacylglycerols, diacylglycerols, monoacylglycerols, free fatty acids, hydrocarbons, and other pigments [56]. Under normal growth conditions, microalgae produce substantial biomass but do not accumulate significant amounts of high-value metabolites such as lipids. Typically, the lipid content of most algal species ranges from 20% to 50% of their dry weight [57].

Fatty acid composition belongs to the key factors in determining the efficacy of biodiesel feedstock. Lipids derived from microalgae exhibit a considerably more diversified fatty acid composition compared to those from plants. Microalgae predominantly produce fatty acids with chain lengths of 12, 16, and 18 carbons; however, certain species can synthesize fatty acids that extend up to 24 carbon atoms. Common fatty acids found in most microalgae species include myristic (C14:0), palmitic (C16:0), stearic (C18:0), oleic (C18:1), linoleic (C18:2), and linolenic (C18:3) acids [58].

While long-chain polyunsaturated fatty acids (PUFAs) are highly valued in the nutraceutical and food industries for their health benefits [59], saturated fatty acids (SFAs) and monounsaturated fatty acids (MUFAs) are preferred for biodiesel production (Figure 1). This preference arises because PUFAs can reduce the stability of the final biodiesel product [60]. High levels of PUFAs, such as C18:2 and C18:3, lower the cetane number, which leads to poor ignition quality [59]. Conversely, short-chain fatty acids with higher proportions of SFAs and MUFAs enhance energy yield, cetane number, and the oxidative and thermal stability of biodiesel [59,61], yet PUFAs, whose content can vary dramatically among different microalgae strains (from less than 3% to nearly 69%), might negatively affect biodiesel quality [59].

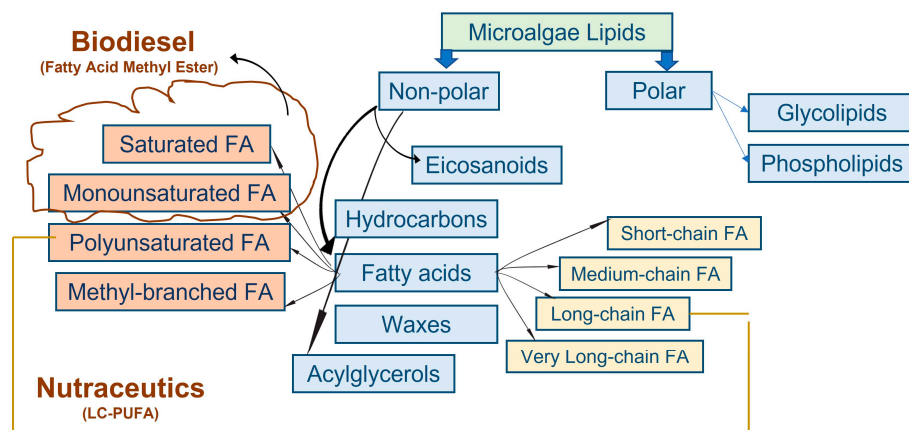


Figure 1. Classification of microalgae lipids and fields of their main application.

The quality of biofuel is primarily determined by the presence of specific fatty acids, including palmitic acid (C16:0), stearic acid (C18:0), oleic acid (C18:1), linoleic acid (C18:2), and linolenic acid (C18:3) [62]. Biodiesel high in oleic acid (C18:1) has been noted for its excellent fuel characteristics, including good ignition quality, oxidative stability, heat of combustion, cold filter plugging point, viscosity, and lubricity [63].

Specific studies have shown varying fatty acid profiles in different microalgae strains: *C. vulgaris* contains 27.32% SFAs and the lowest PUFA content at 38.71%; *C. fusca* holds 28.64% SFAs; *O. submarina* includes 17.17% SFAs with the highest PUFA content at 47.75%; and *Monoraphidium* has 19.01% SFAs [64].

Eight microalgae species isolated and identified from Kota Kinabalu, Sabah, demonstrated commendable lipid content, ranging from 11.69% to 39.00% dry weight, and lipid productivity from 21.11 to 252.64 mg/L/day. Four isolates—*A. falcatus*, *C. emersonii*, *A. obliquus*, and *C. muelleri*—produced more than 80% of C₁₄ to C₁₈ fatty acids. Notably, *C. muelleri* was identified as the optimal candidate for biodiesel production due to its moderate lipid productivity (42.90 mg/L/day), highest lipid content (39% dry weight), high levels of MUFAs and C₁₄–C₁₈ FAs (81.47%), and highest proportion of oleic acid (28.38%), all of which are desirable characteristics for high-quality biodiesel production [65].

Some other algae species also show promise as candidates for biofuel production due to their fatty acid composition. As an example, *Pavlova lutheri*, a microalga popular in aquaculture and a source of high-quality biodiesel feedstock, had the highest total lipid content (313.59 mg/g dry weight) and a substantial amount of fucoxanthin (3.13 mg/g dry weight). It also had the highest level of total n-3 PUFAs (28.01%), including significant amounts of eicosapentaenoic acid (17.76%) and docosahexaenoic acid (7.61%) [66]. The lipids in these microalgae, rich in fucoxanthin and n-3 PUFAs such as eicosapentaenoic acid and docosahexaenoic acid, have great commercial potential as nutraceuticals. *Chaetoceros gracilis* had the highest fucoxanthin content (5.19 mg/g dry weight) and also contained a relatively high level of total lipids (228.87 mg/g dry weight) and 10.67% eicosapentaenoic acid. *Nannochloropsis oculata* contained the greatest amount of eicosapentaenoic acid (26.21%), while *Isochrysis galbana* had the highest level of docosahexaenoic acid (8.76%) [62]. Both species also had significant fucoxanthin content (1.71 and 4.44 mg/g dry weight, respectively).

In summary, fatty acid composition is crucial in determining the efficacy of biodiesel feedstock. Microalgae produce a more diverse range of fatty acids compared to plants, with chain lengths from 12 to 24 carbons. Preferred fatty acids for biodiesel include palmitic (C16:0), stearic (C18:0), oleic (C18:1), linoleic (C18:2), and linolenic (C18:3) acids. Saturated and monounsaturated fatty acids enhance biodiesel stability and performance, while high levels of polyunsaturated fatty acids can reduce stability and cetane number, affecting ignition quality.

3.2. Enhancing Lipid Productivity in Microalgae through Various Manipulation Techniques

Nutritional conditions are expected to significantly influence lipid content and fatty acid composition in both microalgae and seaweeds, which in turn may impact biodiesel properties. The cultivation of *Tetradismus obliquus*, *Heterochlorella luteoviridis*, and *Chlamydomonas reinhardtii* in both unmodified and modified fish farm wastewater revealed that fish farm wastewater nutrients promote the accumulation of desirable biodiesel fatty acids [67]. The primary fatty acid components (C₁₆–C₁₈) constituted up to 95% of all samples, with each containing over 22% SFAs and more than 60% unsaturated fatty acids. When compared to ASTM D6751-02 and EN 14214 biodiesel standards, biodiesel derived from fish farm wastewater exhibited desirable fuel properties, including cetane number, kinematic viscosity, density, higher heating value, cold filter plugging point, and iodine value. *Chlamydomonas reinhardtii* demonstrated particularly suitable biodiesel properties, especially regarding the cold filter plugging point [67].

A similar study conducted in Mexico focused on native microalgae strains typical of that region. Five native strains, namely *Chlorella miniata*, *Coelastrella* sp., *Desmodesmus quadricauda*, *Neochloris oleoabundans*, and *Verrucodesmus verrucosus*, were selected and cultivated using municipal wastewater and foliar fertilizers to evaluate their lipid production capabilities [68]. The highest total lipid concentrations were observed in *Coelastrella* sp. (44–46%), *Verrucodesmus verrucosus* (43–44%), and *Neochloris oleoabundans* (35–37%) [68], which were significantly higher than those reported for microalgae species isolated and identified from Kota Kinabalu, Sabah [65]. An analysis of methyl esters of fatty acids revealed that *Coelastrella* sp. and *V. verrucosus* produced lipids with 82.9% and 91.28% of their fatty acids comprising C₁₆–C₁₈ carbon chains, respectively. Cultivating microalgae on polluted media, such as municipal wastewater treatment plant reservoirs, could potentially support biodiesel feedstock production while also contributing to water purification through bioremediation.

Under conditions of 50% nutritional limitation compared to the recommended levels, notable lipid enrichment was observed in *C. vulgaris*, *O. submarina*, and *Monoraphidium* regarding SFAs, and in *C. fusca* and *O. submarina* concerning MUFAs [64]. This lipid enrichment significantly enhances biodiesel fuel properties. Nutrient limitation, therefore, has a positive impact on the functional properties of biodiesel derived from the tested microalgae strains, as clearly demonstrated in *C. fusca*, *O. submarina*, and *C. vulgaris*.

3.2.1. Nutrient Stress as an Approach to Enhance Lipid Content in Microalgae

The application of nutrient stress to enhance lipid content in microalgae shows variable effectiveness across species and cultivation conditions. Under nutrient stress, however, microalgae alter their fatty acid metabolism towards the synthesis and accumulation of triacylglycerols. For example, studies on *Chlorella* sp. cultured at various temperatures (10–45 °C) under different nutrient-depleted conditions showed substantial lipid yield increases under dual-stress scenarios. Optimum growth was achieved at 35 °C, with lipid yields averaging 55% (*w/w*). The resulting FAMES in the C₁₄ to C₁₆ range were found suitable for biodiesel and low-carbon aviation fuel, emphasizing the potential of *Chlorella* sp. under nutrient deprivation for high-quality biofuel production [69].

Prorocentrum donghaiense has a significantly higher lipid production potential compared to other oleaginous microalgae like *Nannochloropsis gaditana* and *Phaeodactylum tricornutum*, with a lipid content of 49.32% ± 1.99% and productivity of 95.47 ± 0.99 mg/L/day—approximately double that of the comparator species [70]. Similarly, *Microchloropsis salina* [71], *C. muelleri* [65], *M. pusillum*, and *O. multispurus* [34] have shown lipid accumulation exceeding 75% of dry weight under nutrient deficiency, enhancing lipid profiles for biodiesel production.

When nitrogen is scarce in the cultivation medium, microalgae accumulate significant amounts of lipids, but due to the nitrogen deficiency, the cells produce insufficient proteins, leading to reduced biomass production [72]. In other words, while nutrient starvation can increase lipid content per unit biomass, it often results in reduced overall biomass yield,

impacting the total lipid output. For instance, *Ettlia* sp. YC001 under nutrient starvation can increase lipid content by 3–7%, but biomass growth is significantly restricted. Under nutrient-sufficient conditions, lipid productivity can reach up to $6.5 \pm 0.1 \text{ g/m}^2/\text{day}$. High lipid accumulation in biofilms' top layers is attributed to natural nutrient deficiency and intense light exposure [73].

C. vulgaris achieves the largest biomass ($727 \pm 19.64 \text{ mg}\cdot\text{L}^{-1}$) and highest optical density (0.492 ± 0.00) when cultured with 80% salmon farming wastewater. Limited nitrogen access increases lipid content, with the highest lipid content (11.81%) found in the most diluted culture medium (AWW20) with an initial nitrogen content of $6.67 \pm 0.06 \text{ mg}\cdot\text{L}^{-1}$ [74]. *Microchloropsis salina* efficiently utilizes nitrate and organic nitrogen, achieving the highest biomass and lipid productivities (0.11 g/L/day and $52.11 \mu\text{g/mL/day}$, respectively) with yeast extract. Salt stress at 300 mM NaCl enhances lipid production by 80%, despite only a 30% increase in growth, and mixotrophic cultivation with glycerol boosts lipid content to 86.5% of total fatty acids, ideal for biodiesel [71].

Similar results were obtained for *Pseudochlorella pringsheimii*. For scalable biodiesel production, *Pseudochlorella pringsheimii* has proven promising. Cultivated under nutrient-deprived conditions in a 2000 L photobioreactor, the strain yielded high lipid contents of 25% (*w/w*) and a productivity of 74.07 mg/L/day . The subsequent conversion to biodiesel achieved a transesterification rate of $91.54 \pm 1.43\%$, with FAME profiles with C16:0, C18:1, C18:2, and C18:3 as main constituents aligning with ASTM and EU standards [75].

Although comparative studies reveal significant adaptations among different microalgal species under nutrient stress, each presenting unique advantages for biodiesel production, optimal productivity is achieved through a balance, often favoring nutrient-sufficient conditions that harness natural stress factors like light exposure.

3.2.2. Salinity Stress as an Approach to Enhance Lipid Content in Microalgae

Salinity stress is one of the environmental factors that can be easily regulated during microalgae cultivation to enhance lipid accumulation. Various studies have demonstrated the effectiveness of applying salinity stress to different microalgae species, including freshwater species like *Chlorella* sp. and marine species such as *Dunaliella*, *Tetraselmis* sp., and *Nannochloropsis* sp. Salinity stress influences microalgae at the physiological and biochemical levels [46], triggering a series of complex mechanisms that allow the organisms to adapt to adverse conditions. These mechanisms include the accumulation of osmoprotective solutes, production of antioxidant enzymes, regulation of ion exchange processes, and a shift from active cell division to energy storage in the form of lipids [76,77]. In particular, under hyposalinity, *D. tertiolecta*, which endures fluctuating salinity due to factors like global warming, remodels its polar lipid composition, altering membrane fluidity and permeability. These modifications enhance photosynthetic efficiency and cell signaling, revealing this microalga's adaptability to salinity changes [78].

Recent studies have expanded our understanding of these mechanisms. *Dunaliella*, *Tetraselmis* sp., and *Nannochloropsis* sp. are marine microalgae known for their ability to grow in high-salinity environments. Some findings revealed that varying NaCl levels in the cultivation medium led to substantial increases in lipid accumulation in these species. As an example, the growth of *Dunaliella* sp. was significantly higher ($p < 0.05$) and more rapid at 10 ppt compared to both 30 ppt and 50 ppt. Additionally, the protein, lipid, and carbohydrate contents were also higher at 10 ppt [79]. Similarly, high salinity resulted in a reduction in phospholipid and glycolipid content in *Tetraselmis striata*, while upregulating diacylglycerol hydroxymethyltrimethyl- β -alanine betaine lipids at 20°C . Moreover, at higher temperatures, there was an observed accumulation of omega-6 fatty acids, associated with changes in glycolipid composition and a reduction in lipid species esterified to polyunsaturated fatty acids [80]. In opposition, *Scenedesmus* sp. BHU1 exhibited reduced photosynthetic attributes under higher salinity (0.4 M) yet increased lipid content through a two-stage cultivation method [81].

The synergy between salinity stress and other factors such as phytohormones also presents a novel avenue for enhancing lipid production. Yang et al. (2023) [82] explored the influence of phytohormones, particularly abscisic acid, on *Chlorella pyrenoidosa* under saline conditions. Their results highlighted a marked increase in lipid productivity, magnified 3.7-fold, through the combined effects of salinity (20 g/L salt) and phytohormone treatment (20 mg/L abscisic acid). Transcriptome analysis further confirmed that this synergistic approach upregulated genes related to metabolism and photosynthesis, facilitating increased lipid biosynthesis [82]. Also, it has been showed that a two-stage cultivation strategy with varied NaCl concentration treatment (30 mM and 500 mM) applied for freshwater microalgae *Chlorella vulgaris* YH703 resulted in an increased biomass by 1.3 times and a lipid content increase up to 24.5%, compared to the 12.7% in the control group. The highest lipid content was observed in the second stage under 500 mM NaCl [77].

The impact of nitrogen and phosphorus starvation combined with salinity stress on *Monoraphidium braunii* has been studied. The results revealed significant decreases in biomass, productivity, and photosynthetic pigments under nutrient depletion, with maximum values in nutrient-rich control cultures. Low salinity levels (up to 150 mM NaCl) enhanced these parameters, while higher salinity levels (up to 250 mM NaCl) downregulated them. Nutritional limitations and salinity stress significantly increased lipid content and productivity, doubling the control levels and improving fatty acid profiles and biodiesel quality parameters [76].

Recent findings have also highlighted the positive influence of combining a periodic micro-current with NaCl stress during the cultivation of *Chlorella*. A 3 mA periodic micro-current during the growth stage increased biomass by 61.25% to 290.06 mg/L. Under 8 g/L NaCl in the induction stage, lipid content and yield reached 42.11% and 199.32 mg/L, respectively, representing increases of 53.13% and 39.11%. The study also noted a decrease in polyunsaturated fatty acids and an increase in saturated fatty acids, alongside changes in SOD, CAT, ACCase, and DGAT enzyme activities. This synergistic approach facilitated rapid growth and efficient lipid accumulation in *Chlorella*, laying the groundwork for large-scale biofuel production [83].

Despite significant advancements, the primary challenge remains balancing high lipid content with biomass productivity. In some cases, heightened salinity levels often come at a cost of reduced biomass productivity [71,76], a challenge that researchers aim to mitigate through optimizing culture conditions. However, the prospects of utilizing these findings to refine culture conditions are promising for biodiesel production efficiency.

3.2.3. CO₂ Manipulation as an Approach to Enhance Lipid Content in Microalgae

Among various strategies to enhance microbial lipid production, CO₂ manipulation looks like a promising approach. This method offers the dual benefits of enhanced lipid synthesis and CO₂ mitigation, thereby addressing both environmental and energy challenges. A study by Kumar et al. (2014) highlighted the light spectrum and CO₂ concentration as critical factors influencing microalgae growth and lipid accumulation [34]. They found that the combination of red light and 5% CO₂ significantly increased the fatty acid methyl ester (FAME) content in *Micractinium pusillum* and *Ourococcus multispurus*. Under these conditions, the oleic acid fraction ranged between 35 and 37%. This specific illumination also promoted the highest lipid contents, recorded at 20% for *M. pusillum* and 27% for *O. multispurus*, along with substantial lipid productivity (32 and 36 mg L⁻¹ d⁻¹, respectively) [34].

Further research into CO₂ levels for optimal growth found that 6% CO₂ was recommended for *Nannochloropsis salina* [84], which aligns closely with the previous findings [34]. Higher levels, such as 20% or more, led to medium acidification, pigment loss, and growth inhibition. Interestingly, CO₂ fixation capacity and specific lipid production improved markedly when O₂ was removed from the inlet gas. These parameters were further optimized by gradually increasing the CO₂ concentration. However, it was noted that

extremely high CO₂ levels (100%) completely inhibited cell growth, an effect reversed when atmospheric CO₂ levels were reintroduced [84].

Expanding upon the CO₂ manipulation paradigm, Wu et al. (2019) demonstrated that overexpression of Glucose-6-phosphate dehydrogenase (G6PDH) under high CO₂ concentration (0.15%) significantly elevated lipid content and growth rates in *Phaeodactylum tricornutum* [85]. This enhancement was attributed to increased G6PDH activity, higher transcriptional abundance, and increased NADPH production. Conversely, knocking down G6PDH led to reduced growth and lipid accumulation. Additionally, there was a pronounced increase in polyunsaturated fatty acids like eicosapentaenoic acid in G6PDH-overexpressed strains cultivated under high CO₂ conditions [85].

Similarly, increasing CO₂ concentrations enhanced the growth and lipid productivity of *Chlorella vulgaris*. Optimal results were observed during nitrogen depletion at moderate CO₂ levels. However, it became apparent that excessively high CO₂ concentrations were counterproductive, which highlights the necessity for a calibrated CO₂ supply to maximize lipid yields [86].

3.2.4. Genome-Editing Techniques as an Approach to Improve Enhance Lipid Content and Biomass in Microalgae

Genome-editing techniques have significantly enhanced the quality and quantity of products derived from microalgae. Among these techniques, RNA interference (RNAi), zinc-finger nucleases (ZFNs), transcription activator-like effector nucleases (TALENs), and CRISPR/Cas9 have been applied. RNAi, a method used for gene silencing, has proven effective in increasing lipid content in microalgae such as *Phaeodactylum tricornutum* by targeting the nitrate reductase gene, thereby optimizing metabolic pathways for enhanced lipid biosynthesis [87]. ZFNs and TALENs function through protein-directed FokI nuclease cleavage of target DNA sequences. Although there have been successful genome edits using these tools in some plants and insects, their application in microalgae has been less frequent. The example of the application of TALENs is to create ATP:α-D-glucose-1-phosphate adenylyltransferase (AGPL) mutants in *Coccomyxa* sp., which showed significantly higher lipid content compared to their parental strains when grown in diluted minimal media. However, lipid productivity, critical for biofuel production, did not show statistical improvement [88]. While modifications may lead to higher lipid accumulation, they do not always cause greater biomass productivity. Moreover, direct suppression of starch biosynthesis to boost lipid content can be risky because it might result in lower biomass productivity [88], which is counterproductive for biofuel production. Meanwhile, platinum TALENs were used to target the nitrate reductase gene (NoNR) and the acyltransferase gene (LPAT1) in *Nannochloropsis oceanica*, pointing to potential advancements for lipid production improvements [89].

CRISPR/Cas9 stands out for its efficiency, accuracy, and simplicity, employing the RNA-guided Cas9 nuclease to create precise DNA breaks. This system has been successfully applied in various microalgae species, including *Chlamydomonas reinhardtii*, *Nannochloropsis* sp., *Chlorella* sp., and *P. tricornutum* [90]. For example, in *Chlorella vulgaris* FSP-E, CRISPR/Cas9-mediated disruption of the omega-3 fatty acid desaturase (*fad3*) gene resulted in a 46% increase in lipid content [91]. Another application in *Parachlorella kessleri* involved the gene disruption of AATPL1, which led to more than 30% higher lipid productivity compared to wild-type strains, demonstrating the method's potential for improving biofuel feasibility [92]. In AATPL1-knockout strains, lipid productivity (g L⁻¹ day⁻¹) was measured at 0.23 ± 0.01, 0.15 ± 0.01, 0.12 ± 0.01, and 0.11 ± 0.01 under day lengths of 24, 16, 12, and 8 h, respectively. These values represent increases of 3.3%, 43%, 22%, and 29% compared to the wild-type strain [92].

Additionally, using a DNA-free CRISPR/Cas9 RNP method, research on *Tetraselmis* sp. led to a significant enhancement in lipid content—up to 3.1-fold (21.1% and 24.1% of DCW)—under nitrogen starvation conditions compared to wild-type strains (7.68% of

DCW) [93]. These advancements underscore the promise of CRISPR/Cas9 as a standard and reliable genome-editing tool for various microalgae.

4. Physicochemical Properties of Algae Biodiesel

Most structural lipids in microalgae are naturally long-chain and contain more than two double bonds, which adversely affect fuel quality by yielding a lower cetane number, reduced combustibility, and higher iodine values. Consequently, biodiesel production methods should focus on extraction techniques that selectively enrich neutral lipids such as triglycerides or TAGs from the cellular matrix while minimizing the co-extraction of less desirable structural lipids, such as phospholipids and glycolipids [19].

4.1. Kinematic Viscosity

Kinematic viscosity is an important property of algae biofuel, significantly impacting engine performance and emission characteristics. The proper management of kinematic viscosity within standardized limits is essential for ensuring high-quality combustion and preventing engine deposits. International standards, specifically the American Society for Testing Materials (ASTM D6751) and the European standard (EN 14214) [94], mandate that the kinematic viscosity of biodiesel should be in the ranges of 1.9–6.0 mm²/s and 3.5–5.0 mm²/s, respectively. Meanwhile, the kinematic viscosity limits specified in the automotive fuel standard EN 590 range from 2.00 to 4.50 mm²/s, and the density limits range from 820 to 845 kg/m³ (EN 590, 2013) [95]. For methyl esters, the EN 14214 standard mandates a kinematic viscosity range of 3.50 to 5.00 mm²/s and a density range of 860 to 900 kg/m³ (EN 14214, 2012). Exceeding these limits can lead to increased droplet sizes, a narrower injection spray angle, poorer vaporization, and greater in-cylinder penetration of the fuel spray, which collectively impair combustion performance, elevate emissions, and increase oil dilution [96–98]. Conversely, lower-viscosity fuels may result in higher leakage losses and excessive wear of fuel pumps [99]. Generally, the viscosity of biodiesel varies with the fatty acid composition and is higher than that of conventional diesel [100], and this increase in viscosity is influenced by the methyl ester composition of the biodiesel as well as the degree of unsaturation and the temperature [99]. Biodiesel derived from microalgae such as *Nannochloropsis salina*, *Spirulina platensis*, and *Scenedesmus* exhibits different stability and viscosity profiles [101]. Generally, higher viscosity correlates with species rich in long-chain polyunsaturated fatty acids (LC-PUFAs), such as eicosapentaenoic acid and docosahexaenoic acid [102]. *Spirulina platensis*, with a high palmitic acid content, tends to have poor cold properties but remains within acceptable viscosity ranges at standard temperatures. The viscosity of biodiesel decreases as the concentration of methyl oleate and the degree of unsaturation decrease.

Different algae species produce biofuels with varying kinematic viscosities. For instance, certain species like *Chlorella emersonii* (1.01 mm²/s), *Chlamydomonas reinhardtii* (1.3 mm²/s), *Amphora* sp. (1.13 mm²/s), and *Dunaliella salina* (0.89 mm²/s) have relatively low kinematic viscosities, close to conventional diesel (ASTM). Notably, the NT8a strain of *Chlorella* sp. has been selected from 17 microalgae strains for jet fuel production based on several relevant jet fuel standards [9]. On the other hand, *Nostocales* and some species from *Chlorellales* exhibit much higher kinematic viscosities, sometimes exceeding three times those of the previously mentioned species. As an example, among the 15 algae species [98], *Nannochloris* sp. had the highest kinematic viscosity, measuring 3.73 mm²/s. This variability is attributed to the lipid composition, particularly the presence of long-chain polyunsaturated fatty acids (LC-PUFAs) like eicosapentaenoic acid and docosahexaenoic acid [102]. Importantly, the removal of 50 to 80% of LC-PUFAs from *Nannochloropsis*-based methyl esters can bring the viscosity within current specifications, enhancing oxidative stability [102].

Although some findings showed a low level of viscosity for *Chlorella*-produced biodiesel, some of them disclose opposite data. As an example, *Chlorella vulgaris*-produced

biodiesel has a viscosity value as high as $3.7 = 4.21$ at $40\text{ }^{\circ}\text{C}$ [103,104]. Viscosity has also shown an average correlation of 0.52 with monounsaturated fatty acid content [98].

Research on the density and viscosity of microalgae biodiesel derived from *Nannochloropsis salina*, *Spirulina platensis*, and *Scenedesmus* across broad pressure and temperature ranges revealed that both *Nannochloropsis salina* and *Scenedesmus* biodiesel exhibited poor stability. In contrast, *Spirulina* biodiesel demonstrated high stability, with density and viscosity values that support its potential for practical application.

For *Spirulina platensis*, the fatty acid profile, dominated by palmitic acid (42.25 wt%), suggests that biodiesel from this strain might have poor cold properties. The high amount of palmitic acid is a limitation for its use in engines, as it leads to poorer cold flow properties. However, the viscosity of *Spirulina* biodiesel, measured at $40\text{ }^{\circ}\text{C}$ and 1 bar, was $5.2452\text{ mm}^2/\text{s}$, which falls within the EN 14214 standard range, making it a promising biofuel candidate.

The analysis of the fatty acid methyl ester (FAME) profile for *Scenedesmus almeriensis* biodiesel showed a high presence of palmitic acid (47.12%) and methyl linolenate (18:3), indicating susceptibility to oxidation. Despite good reproducibility, the viscosity value recorded at $40\text{ }^{\circ}\text{C}$ was $7.8417\text{ mm}^2/\text{s}$, exceeding the recommended limits for biodiesel, largely due to its lower purity. This suggests potential challenges in scaling up production for industrial applications [101].

When the viscosity of algal-based biodiesel differs significantly from that of conventional fossil fuel, the density of biodiesel is typically only slightly higher than that of diesel. This slight increase in density results in larger droplet sizes during fuel atomization in the engine, preventing the fuel from burning completely and leading to higher emissions of non-burned hydrocarbons, CO, CO₂, and NO_x [105].

4.2. Surface Tension

Surface tension is an important physical property of fuels that significantly influences their behavior in various applications, including biodiesel. Surface tension plays an important role in processes that involve liquid interfaces, such as the emulsification and extraction of biodiesel from microalgae. In the context of fuel combustion, higher surface tension is generally an obstacle because it resists the formation of smaller droplets. The creation of finer droplets is essential for good combustion in diesel engines as it ensures better mixing with air, leading to more efficient and cleaner burning. Compared to pure biodiesels, binary blending with larger proportions of diesel may lead to a reduction in viscosity, density, and surface tension of the fuel, thereby supporting optimal atomization characteristics suitable for diesel engines [106]. For example, a blend of 20% sunflower oil-based biodiesel and 80% diesel exhibits spray parameters that are fairly close to those of conventional diesel [107].

Biodiesel, derived from FAMES, exhibits unique surface tension properties influenced by the fatty acid composition. At a standard temperature of $20\text{ }^{\circ}\text{C}$, biodiesel typically has a higher surface tension than diesel. For instance, biodiesels produced through transesterification of various vegetable oils exhibit the following average values: castor oil methyl ester at $35.66 \pm 0.06\text{ mN/m}$, neem oil methyl ester at $31.68 \pm 0.05\text{ mN/m}$, and sunflower oil methyl ester at $26.42 \pm 0.05\text{ mN/m}$, compared to diesel's $29.46 \pm 0.06\text{ mN/m}$ [107]. This increased surface tension contributes to larger value droplet sizes in the fuel spray, which tends to reduce the combustion efficiency and increase emissions. The surface tension of biodiesel has been proven to decrease linearly with an increase in temperature, albeit at different rates for different types of biodiesel. For example, sunflower oil methyl ester's surface tension drops below that of diesel at elevated temperatures, enhancing its atomization properties. However, the inherent higher viscosity and surface tension of biodiesels usually mean larger droplet sizes and a narrower spray cone angle compared to diesel [107]. Unfortunately, this consumption mainly pertains to plant-based biodiesel, while information on algae-based biodiesel remains scarce.

The surface tension not only affects the spray characteristics but also plays a role in the combustion regime. For instance, it has been shown that an increase in surface

tension raises the Sauter Mean Diameter (SMD) of biodiesel droplets, leading to less fuel burning in the premixed combustion regime and more in the diffusion combustion regime, contributing to increased NO_x emissions. Higher surface tension can also result in less homogeneous combustion, increasing particle emissions due to incomplete burning, which is associated with soot formation. Experiments conducted on diesel engines using various biodiesel blends reveal the influences of surface tension on engine performance and emissions. In one study, engine tests at 3670 rpm using pure cottonseed biodiesel (CS-B100), emulsified cottonseed biodiesel with water and an emulsifier (CS-E20), and emulsified biodiesel containing *Chlorella vulgaris* cells (CS-ME20) showed significant differences. The brake-specific fuel consumption increased by 41% when using emulsified water fuels (CS-E20) compared to CS-B100, reflecting the impact of increased surface tension and viscosity. Moreover, emissions such as NO_x and CO₂ were significantly lower with CS-ME20, attributed to the improved heating value from *Chlorella vulgaris* cells [108].

The surface properties of microalgae, crucial for biodiesel extraction, are influenced by several factors. One example is *Dunaliella tertiolecta*, where the critical interfacial tension of the cells depends on cell age. Specifically, for cells in the stationary phase at negatively charged interfaces, a decrease in critical interfacial tension by 383 mJ/m² corresponds to a loss of cell hydrophobicity, impacting the efficiency of solvent extraction processes [109]. To enhance the extraction of cell contents, including lipids necessary for biodiesel production, surfactants are commonly used. Cationic surfactants like cetyltrimethylammonium bromide, myristyltrimethylammonium bromide, and dodecyltrimethylammonium bromide significantly lower surface tension through electrostatic interactions, facilitating lipid extraction. The highest FAME recovery from *Scenedesmus* biomass was achieved from high-lipid biomass treated with myristyltrimethylammonium bromide and 3-(decyldimethylammonio)-propanesulfonate inner salt surfactants using a mixed solvent (hexane/isopropanol in a 1:1 ratio) vortexed for only one minute, yielding up to 160 times more FAMEs compared to untreated biomass [110]. The critical micelle concentration of surfactants is important for extraction performance, but the biomass growth stage has a greater impact on surfactant efficiency in disrupting cells and improving lipid extraction.

Anionic surfactants like sodium dodecyl sulfate also enhance lipid solubility and extraction. Their high hydration capacity and ability to lower microalgae cell surface energy increase permeability. For example, incorporating sodium dodecyl sulfate in H₂SO₄-catalyzed hot water extraction significantly boosts lipid extraction from *Chlorella vulgaris* and *Spirulina platensis*, optimizing algal oil conversion to FAMEs [111,112]. Moreover, sodium dodecyl sulfate in H₂SO₄ achieved a maximum FAME yield of 98.3% at 20% moisture in *Nannochloropsis oculata* algae. Additionally, a small amount of water in the biomass or methanol increased lipid extraction efficiency, enhancing FAME yield rather than inhibiting it [113].

Nonionic surfactants provide a milder extraction alternative without affecting the catalytic activity of extracted proteins. Techniques like Cloud Point Extraction have shown the effectiveness of nonionic surfactants like Triton X-114 in phase separation and extraction processes, optimizing biodiesel production. This method was validated using in situ extraction of microalgae with an *Acutodesmus obliquus* culture [114].

5. Microalgae as the Sustainable Feedstock for Sustainable Aviation Fuel and Biohydrogen

Microalgae have garnered significant scientific interest as a promising feedstock for biodiesel and biohydrogen production, given their high lipid content and rapid growth rates, as outlined in the previous sections. This green biomass offers a sustainable alternative to conventional fossil fuels, addressing both energy security and environmental concerns (Figure 2). The cultivation of microalgae requires non-arable land and can utilize saline or wastewater, mitigating the impact on food and freshwater resources. The biochemical composition of microalgae, predominantly characterized by fatty acids and polysaccharides, makes them apt for conversion into biodiesel through transesterification and

biohydrogen via biological or thermochemical processes. However, the high costs associated with microalgae cultivation, processing, and conversion present significant challenges to commercialization. Advances in biotechnological methods and process engineering are essential to enhance the economic feasibility and scalability of algae-based biofuels.

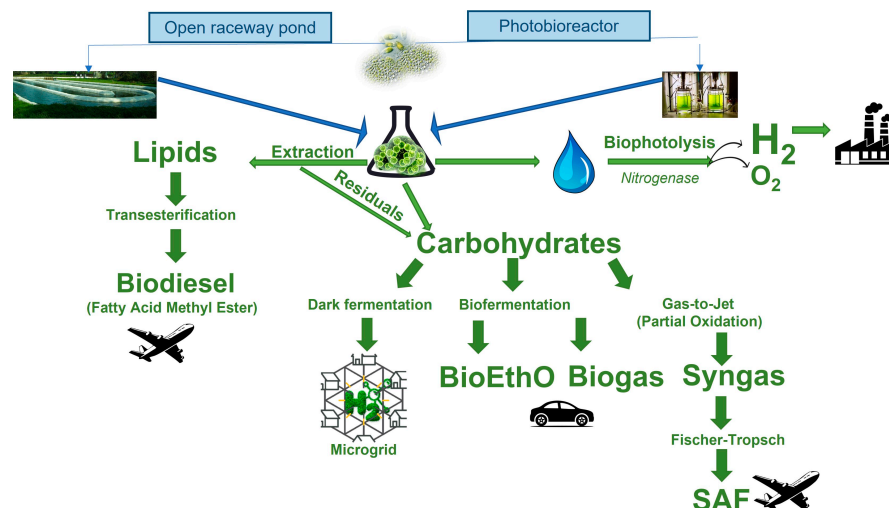


Figure 2. Possible pathway for biofuel production from algae.

5.1. Biodiesel and Sustainable Aviation Fuel

The most widely utilized technology for producing low-carbon fuels, such as aviation biofuel from microalgae feedstock, is the oil-to-jet process [115] (Figure 2). This involves converting microalgae oil into hydrocarbons with carbon chain lengths ranging from 8 to 16.

Currently, the biofuels approved by ASTM as sustainable aviation fuels include hydroprocessed esters and fatty acids (HEFA), Fischer–Tropsch synthesis (FT), alcohol-to-jet (ATJ), and direct sugar to hydrocarbon (DSHC). Among these, only HEFA has advanced to Technology Readiness Level (TRL) 7–8. It has been shown that the modeled algae-to-HEFA pathway has the potential to achieve a minimum fuel selling price as low as USD 4.70 per gasoline gallon equivalent, in tight relation with achieving significant carbon intensity reductions [116].

Despite the numerous advantages of algae as a feedstock for sustainable aviation fuel (SAF) outlined in the previous sections of this review (see Sections 2 and 3), algal lipids are not commonly utilized for commercial fuel production. This is primarily due to the high costs associated with algae production, which involve costly cultivation methods such as open raceway ponds, significant energy input, and relatively low baseline levels of lipid productivity and fatty acid composition required for economically feasible biofuels. However, ongoing research efforts have made substantial progress in addressing these challenges. Some researchers have demonstrated advancements in lipid productivity, combining biomass productivity and lipid content, achieving ranges of 7–9 g m^{−2} per day [41,73] (see Section 3.2). Despite these advancements, large-scale production of biofuel from microalgae has not yet been realized. Nevertheless, there are several projects worldwide aimed at reaching Technology Readiness Level 5, indicating a move towards scaling up this promising technology.

Examining efforts in the U.S., China, Germany, France, and other regions regarding biodiesel production from microalgae reveals different levels of success, technological advancements, and policy impacts. The U.S. government, through the Department of Energy’s Bioenergy Technologies Office (BETO), is advancing algal biofuel development to bolster energy security and curtail greenhouse gas emissions. BETO employs an integrated modeling framework with resource assessments, techno-economic analysis, and life-cycle evaluations to scrutinize the cultivation and conversion of algal biomass. Although microalgae productivity estimates remain conservative, BETO aims to produce 5 billion gallons of

renewable algae diesel annually by 2030 while significantly reducing carbon emissions. Key advancements include design pathways producing biofuel at less than USD 5 per gallon (gasoline equivalent), exemplifying a rigorous balanced approach prioritizing sustainability and economic feasibility [117]. Achieving these targets, however, requires continual research and development, substantial investment, and overcoming technological and infrastructural challenges. In China, despite substantial government support, rapid growth in microalgae-based biodiesel remains challenging. Research identifies major obstacles: high production costs (2.29 USD/kg compared to USD 3.4 per gallon diesel [118]), limited microalgae productivity, and few operating days per year. Economic feasibility is compromised by inadequate policy support and technical inefficiencies. Sensitivity analyses point to crucial economic factors like productivity enhancements, extended operation periods, and improved extraction efficiencies as transformative. Sun et al. (2019) advocate for policy restructuring to foster development [119]. Additionally, studies by Yang et al. (2015) show that socio-economic benefits from algae biodiesel vary regionally. Urban areas like Yunnan and Guangxi could gain significantly, indicating that targeted regional policies might enhance biodiesel's impact [120].

German and French efforts reflect a technology-centric approach to microalgae biofuel production in Europe. Projects such as the IGV GmbH-bioalgastral SAS initiative on La Reunion emphasize photobioreactor technologies that enhance biomass productivity while reducing costs. In glass photobioreactors that were installed on La Reunion by IGV in 2011–2014, it was planned to produce biofuels from microalgae with a total volume of 82,000 L in 2012 [121]. In parallel, the Technology University of Munich has implemented a state-of-the-art 1500-square-meter Algae Tech Facility designed to simulate global climate conditions, enabling the cultivation of diverse algal strains [122]. The collaboration with Airbus highlights the aviation industry's interest in bio-kerosene, exploring closed-loop saltwater systems for biofuel production. This facility, part of the broader "AlgenFlugKraft" project, consolidates cutting-edge research and application with substantial institutional support and industry collaboration, reflecting a robust framework for sustainable development.

The FUELGAE consortium, comprising 13 partners, focuses on developing advanced liquid aviation and maritime fuels. This project revolves around the innovative catalytic conversion of biocrude and lipids into superior fuels. The primary goal is to establish a novel method for producing advanced liquid fuels utilizing various CO₂ emission streams from two different industrial sectors. To achieve this, a pilot photobioreactor is set to be integrated into their infrastructure, utilizing selected microalgae strains. The project aims to advance the technology for algae-based biofuel production to Technology Readiness Level 5 (TRL5) [123]. The integrated sustainability assessment approach, as seen in the D-FACTORY project, further supports optimizing microalgae cultivation and processing using *Dunaliella* as the model, reinforcing the EU's commitment to pioneering green technologies [124].

Alternatively, carbohydrate-based pathways, including sugar-to-jet, alcohol-to-jet, and gas-to-jet processes, form the second group of biofuel conversion methods. For instance, the sugar-to-jet process involves the biochemical conversion or fermentation of biomass sugars into bio-jet fuel using enzymatic catalysts. Although promising, alternative methods like sugar-to-jet and gas-to-jet conversions are still in their early stages [115]. Gasification is the initial step in the gas-to-jet technology, where biomass is converted into syngas. This syngas reacts with hydrogen in the presence of a catalyst, typically cobalt or iron, through Fischer–Tropsch synthesis to generate liquid hydrocarbon fuels.

Although microalgae contain a relatively high water content, which can decrease the efficiency of gasification and produce low-energy syngas, a drying process is essential to reduce moisture before gasification. During gasification, biomass undergoes partial oxidation at high temperatures (800–1000 °C), producing syngas with significant proportions of H₂ (6–55%) and CO (8–53%). For instance, a temperature of 800 °C was found to be optimal for the gasification of *Chlorella vulgaris*, yielding the highest H₂ (4.53 mmol/g biomass with a purity of 34.62 vol%) and the highest generation peak intensity of 0.14 mmol/min/g. Conversely, the highest lower heating value (LHV) (8.92 MJ/Nm³), total gas yield (73.54 wt%),

and carbon conversion efficiency (49.54%) were achieved at 900 °C [125]. Similarly, an operating range of 700–850 °C was ideal for the gasification of *Nannochloropsis oculata* and *Dunaliella salina* [126].

The Ni catalyst was identified as the most effective for maximizing H₂ yield (8.22 mmol/g biomass with 56.12 vol% purity) and achieving the highest total gas yield of 80.21 wt% [125]. Additionally, the steam-to-biomass ratio is crucial and should be maintained at around 0.4–0.6 [126]. Higher temperatures were also highlighted as important for the quality of syngas produced from *Cladophora glomerata*; as the gasification temperature increased from 700 °C to 900 °C, tar content in the gas significantly decreased, while hydrogen yield increased. Increasing the steam-to-biomass ratio further enhanced hydrogen yield and tar destruction [127].

However, the syngas produced from the gasification of microalgae biomass generally has a relatively low calorific value, only sometimes exceeding 13 MJ/kg, as demonstrated for *Chlorella vulgaris* [128]. To enhance the calorific value, the syngas was further upgraded via the Fischer–Tropsch process to produce FT bio-jet fuel [129]. The integration of Power-to-X via methanol (MTJ) and FT routes with Biomass-to-X (BtX) through an algae-based biorefinery forms a Power- and Biomass-to-X (PBtX) process. This includes algae remnant utilization for H₂/CO₂ production, wastewater recycling, and heat integration. Economic optimization suggests a PBtX process, yielding fuels at 211 EUR/MWhLHV, a reduction of up to 21% in costs compared to stand-alone methods. High algae costs (>40 EUR/t) or MTJ costs (2000–2400 EUR/t_{jet}) affect economic performance [130].

FT fuel production requires a catalyst to convert syngas into long chains of liquid hydrocarbon fuel. Recent research has demonstrated that modified algae-derived biochar can serve as an effective support for cobalt catalysts in Fischer–Tropsch synthesis. Specifically, the *Cladophora glomerata* algae biochar-supported cobalt catalyst (15%) achieved the highest FT rate of 0.345 g HC/(g cat. h) and a CO conversion rate of 67%, surpassing the performance of industrial γ -Alumina-supported catalysts. Although the selectivity for C₅+ liquid hydrocarbons showed a slight decrease, the results indicate that biochars from *Cladophora glomerata* algae compared to rice husk and coconut shell are the most appropriate promising alternatives to γ -Alumina for cobalt catalyst support [131]. Furthermore, high-porosity modified biochars from pyrolyzed peanut shell and *Cladophora glomerata* algae, when used as a support for potassium-promoted cobalt catalysts, proved to be highly effective. The catalysts exhibited significant activity and selectivity improvements, with α -olefin selectivities of 38.67% and 35.49% for C₂–C₁₃ hydrocarbons, respectively [132]. These advancements point to the potential of biochars in enhancing FTS and promoting sustainable fuel production.

Although challenges remain, algae may be a cost-effective alternative to fossil fuels in aviation. Reports suggest that achieving a 30% replacement from algal sources would require 108 billion liters, corresponding to a land area of 11,345 km² [133,134]. This land requirement is likely more feasible than using traditional plant-based feedstocks for biofuel production. While current advancements indicate potential, the scalability and economic feasibility of algae biofuels will be key determinants of their future in decarbonizing the aviation sector.

5.2. Biohydrogen

The development of biohydrogen as an alternative energy source offers significant economic and environmental benefits. Biohydrogen, a form of green hydrogen, achieves negative CO₂ emissions, unlike electrolytically produced green hydrogen. For instance, it has been demonstrated that when cultivated for biohydrogen production, 1 L of *Chlorella kessleri* microalgae suspension (with an initial optical density of 0.798) absorbs an average of 0.195 ± 0.001 g of CO₂ per day [135]. Additionally, biohydrogen has the highest gravimetric energy of any fuel at 120 MJ/kgH₂, can be transported using existing pipelines, and produces only water upon combustion, making it a sustainable and ecologically sound energy source [136].

Hydrogen production from microalgae presents itself as a sustainable method of energy generation that can help mitigate fuel shortages while recycling waste [137]. After oil is extracted from algae, which can be used for sustainable aviation fuel (see Section 5.1), about 50% of algae residue from the initial biomass remains [138]. Current high-value treatments for the sustainable utilization of this algae residue include the production of dark-fermentation H_2 , biomethane, and other gaseous biofuels [139,140].

Hydrogen production yields from microalgae residues have shown significant promise. For example, 192.35 mL/g volatile solid dark-fermentation H_2 and 183.02 mL/g volatile solid dark-fermentation H_2 were obtained from the lipid-extracted biomass of *Dunaliella primolecta* and *Dunaliella tertiolacta*, respectively, within 19 h at a biomass concentration of 2.5 g/L [141]. Additionally, hydrogen yields range from 0.37 to 19 mL H_2 /g volatile solids, with the highest yields reported from *Chlorella vulgaris*, followed by lipid-extracted *Scenedesmus* sp. [142]. Also, biohydrogen was produced in a laboratory from locally available *Euglena acus* microalgae in Bangladesh, yielding approximately 3 to 5 mL of biohydrogen for every 40 mL microalgae suspension, depending on concentration [143].

The current cost of producing biohydrogen ranges from 1.15 to 9.65 EUR/kg H_2 , whereas the cost of producing green hydrogen via electrolysis varies between 2.51 and 11.94 EUR/kg H_2 [144]. Economic assessments reveal that the cost of producing biohydrogen largely depends on the design of the bioreactor and the production pathway [145]. As an example, the specific cost of photofermentative hydrogen was estimated using a 20 L pilot-scale photobioreactor with *Rhodobacter capsulatus* YO3 (hup). Annual capital and operating costs were USD 124 and USD 42, respectively. The hydrogen production cost was 2.7 USD/mol (1362 USD/kg H_2), decreasing to 395 USD/kg H_2 with equipment subsidies. Utilizing wastewater feedstock and increasing hydrogen productivity can further lower costs [146]. Additionally, the cultivation of *Galdieria sulphuraria* microalgae has demonstrated that techno-economic efficiency varies with the configuration of the reactor. A confined pond and a helical photobioreactor were used for cultivation; the photobioreactor yielded a biomass productivity of 1.575 kg m^{-3} /day and a substrate yield of 0.57 g g^{-1} at a minimum cost of 2869 USD/DW metric ton [147].

One of the primary obstacles in commercializing microalgae-based hydrogen production lies in inadequate process engineering, a lack of knowledge in strain enhancement, outdated technologies for microalgae biomass cultivation and harvesting, and low productivity rates [137,145]. Gene-editing technology holds the potential to overcome some of these challenges; however, the stability and safety of genetically modified microalgae remain crucial considerations [137]. For instance, the susceptibility of key fermentation enzymes to oxygen can further complicate the development of economically viable microalgal biohydrogen production systems. Specifically, genetic modifications to the hydrogenase diaphorase A (HYDA) of *Chlorella* sp.—including introducing changes to the active site of the enzyme at amino acid residues A105I, G113I, V265W, and V273I—have demonstrated significant increases in biohydrogen production, achieving up to 30 times more than the wild type when exposed to oxygen [148]. Additionally, for downstream biohydrogen production, reducing the demand for light and exploring more energy-efficient dark fermentation strategies are essential [137].

Hydrogen production methods encompass several biological mechanisms, with biophotocatalytic and fermentation processes being the most prominent [145] (Figure 3). In addition to these traditional methods, biohydrogen can also be produced from microalgae using electrochemical methods such as microbial fuel cells, microbial photoelectrochemical cells, and microbial electrolysis cells. These electrochemical methods are frequently preferred over typical chemical, biological, and biochemical processes due to their extensive application potentials and efficiency [149]. The process of biophotolysis in microalgae involves the coordination of chlorophyll-containing reaction centers with hydrogenase and nitrogenase enzymes. Direct photolysis of water under anaerobic conditions generates hydrogen and oxygen gases. This process can be categorized into two mechanisms: direct and indirect biophotolysis, both of which are analogous to photosynthesis. Notably, green

microalgae are capable of conducting biophotolysis in the absence of oxygen, setting them apart from other microorganisms [145,150]. Direct biophotolysis is a two-step reaction. In the first stage, water molecules are split into protons, electrons, and oxygen by Photosystem II. In the second stage, hydrogen is produced with the aid of hydrogenase enzymes [149].

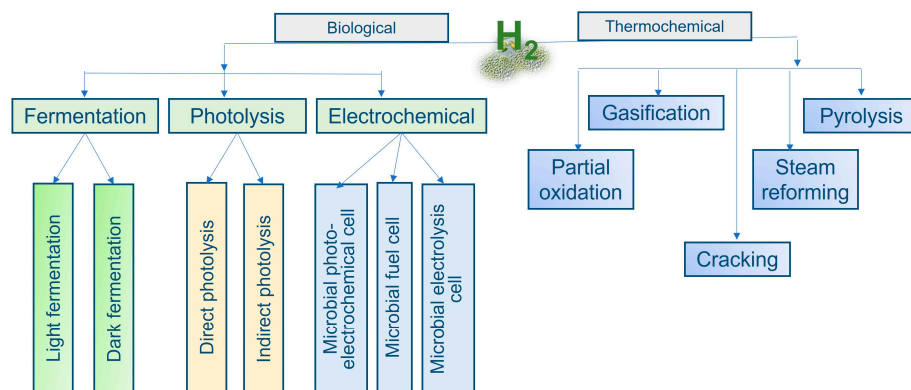


Figure 3. Biological and thermochemical methods of biohydrogen production.

Despite the high theoretical efficiency of direct biophotolysis, practical conversion rates remain low due to limitations in microalgal development and the photosynthetic mechanism. In contrast, dark fermentation involves the microbial conversion of substrates like carbohydrates and proteins into hydrogen. This method allows for the simultaneous production of biohydrogen and valuable by-products like acetic, lactic, and butyric acids [145,151].

Physical and chemical pretreatments such as mechanical, heat, ultrasonic, acid, base, and ozonation have been widely applied to disrupt and disintegrate the cell wall of microalgal biomass, enhancing the subsequent biological conversion process [142]. Hydrogen yields from algal biomass through dark fermentation can be significantly increased to 50–70% of their theoretical values by employing effective pretreatments and optimizing the biomass carbon/nitrogen ratios, using domesticated hydrogen-producing bacteria as the inoculum [152]. For instance, *Chlorella kessleri*, when cultivated, has shown the ability to purify wastewater with high efficiency, further increasing the yield of biohydrogen with the addition of starch [135]. Additionally, subsequent processes like photofermentation and anaerobic digestion can boost the total energy yield to 16.2 kJ/g VS [152].

Integrating biohydrogen into microgrids presents a promising opportunity to achieve sustainable goals and effectively mitigate global climate change. This strategy is particularly noteworthy as it aligns with the growing need for renewable energy sources to replace fossil fuels in power generation systems [153]. Although significant advancements have been made in understanding the operational parameters and scalable improvements for biohydrogen production pathways, there is a lack of studies on the integration of biohydrogen into green hydrogen microgrids. While almost all investigations focus on electrolytically produced green hydrogen, some findings emphasize that the integration of microgrids in chemical industries and the use of purified hydrogen steam to produce electricity in a PEM fuel cell are not yet economically viable due to the long discounted payback period [154]. Meanwhile, there has been a notable focus on the economic viability of biohydrogen production from biomass, particularly from microalgae within a circular economy framework, ensuring that biomass supply chains are effective and sustainable [155].

Biohydrogen's potential in microgrids is significant, particularly for rural and remote areas that require resilient and reliable energy sources. Microgrids powered by biohydrogen can serve these communities by utilizing local biomass waste residues and microalgae biomass cultivated in wastewater treatment facilities, thereby promoting energy independence and sustainability [156]. Furthermore, biohydrogen can be used in fuel cells for instantaneous power generation, offering energy resource diversification and enhancing energy resiliency. Research indicates that overcoming the technical and economic barriers

of biohydrogen production could enable it to capture a significant share of the global power generation market. This shift could support global efforts towards achieving Net Zero Emissions by 2050, highlighting biohydrogen's critical role in the future of green energy and sustainable microgrids [153].

In conclusion, microalgae-derived biohydrogen represents a promising and sustainable energy source with numerous environmental benefits. While significant strides have been made in optimizing production methods and reducing costs, further research and technological advancements are critical to overcoming existing challenges and realizing the full potential of this renewable energy resource.

6. Conclusions and Perspectives

Microalgae demonstrate significant potential as a sustainable and renewable resource for biofuel production due to their high lipid content and rapid growth rates. Among the wide range of microalgae strains, freshwater species like *Chlorella* sp. and marine species such as *Dunaliella*, *Tetraselmis* sp., and *Nannochloropsis* sp. are the most commonly utilized for biofuel production. They present a promising solution to global energy needs, beyond just biodiesel, including sustainable aviation fuel and biohydrogen.

The cultivation of microalgae can be conducted using various methods such as photobioreactors and open pond systems, each with its own unique benefits and challenges. While photobioreactors offer controlled conditions that enhance yield, they are often capital-intensive. Conversely, open pond systems are cost-effective but more susceptible to environmental variables. Hybrid systems that combine the strengths of both approaches represent a promising pathway for scalable biofuel production.

Biodiesel is currently the primary alternative to diesel fuel in the European Union, accounting for 73% of total biofuel production [157] and still growing. Although Germany, France, and Spain host the largest number of microalgae producers in Europe [23], the application of microalgae for biodiesel remains relatively low. Although some R&D projects supported by governmental and private grant institutions such as the European Commission and the U.S. Department of Energy's Bioenergy Technologies Office have shown promising outcomes [158], widespread adoption is still in the initial stage.

Multiple barriers have hindered the rapid growth of microalgae-based biodiesel, the most significant being high production costs and limited microalgae productivity. When considering the EU Directive 2003/30, which aims to promote the use of biofuels, and the fact that Asia produced 20 billion liters of biodiesel in 2022 [159], it becomes evident that third-generation biofuels derived from algae must be developed. This development is critical not only for alleviating competition for biomass but also for addressing ecological issues such as deforestation and biodiversity loss. Future research and development efforts should focus on further optimizing cultivation conditions, enhancing genetic engineering techniques, and developing cost-effective harvesting and processing methods. Integrating microalgae cultivation within existing industrial and agricultural practices can also provide synergistic benefits. Additionally, improving policy frameworks and financial incentives will be crucial for scaling up production.

Sustainable aviation fuel derived from microalgae represents a significant opportunity for decarbonizing the aviation sector. The most widely utilized technology for producing sustainable aviation fuel from microalgae feedstock is the hydroprocessed esters and fatty acids (HEFA) pathway, which involves converting microalgae oil into hydrocarbons with carbon chain lengths ranging from 8 to 16. This pathway has reached Technology Readiness Level (TRL) 7–8, indicating its near-commercial maturity. However, the high costs associated with algae production, including cultivation, harvesting, and processing, have limited the widespread adoption of algal lipids for commercial fuel production. To overcome economic and technological barriers, several strategies have been proposed:

1. **Cultivation Optimization:** Tailoring microalgal cultivation conditions to optimize the production of desirable fatty acids can enhance fuel properties, while genetic engineering technologies such as CRISPR/Cas9 offer precision in modifying lipid

biosynthetic pathways for higher yields and better fuel characteristics. Employing hybrid cultivation systems to balance cost and productivity can also optimize resource use.

2. **Advanced Harvesting Techniques:** Developing energy-efficient harvesting methods, such as enhanced flocculation, co-cultivation with other microorganisms, and the use of biofilm reactors to reduce costs.
3. **Resource Efficiency:** Integrating microalgae cultivation with wastewater treatment facilities to maximize the use of nutrient-dense waste streams, hence decreasing operational expenses and overall environmental footprint.
4. **Policy Support and Incentives:** Government subsidies, tax incentives, and other financial support are essential to making SAF from microalgae commercially viable. Most bio-jet fuel production technologies incur costs that are at least 120% higher than conventional fossil-based jet fuel, while achieving emissions reductions of at least 27%. Despite these high costs, only 38% of existing policies provide monetary incentives to SAF producers, resulting in SAF production operating at only 3.5% of its total potential capacity [160].

Biohydrogen production from microalgae offers significant economic and environmental benefits, achieving negative CO₂ emissions and providing a sustainable hydrogen source. Hydrogen from microalgae can be produced through biophotolysis, where sunlight and water are used to generate hydrogen and oxygen, and through dark fermentation of microalgae residues after lipid extraction. However, biohydrogen production faces several challenges, including low hydrogen yields, high production costs, and the need for process optimization. Potential solutions harmonize in some points with those proposed for sustainable aviation fuel and include the following:

1. **Genetic Modifications:** Enhancing microalgal strains through genetic engineering to increase hydrogenase activity and overall hydrogen yield.
2. **Bioreactor Design:** Developing advanced bioreactors that maximize sunlight capture and gas exchange and optimize growth conditions.
3. **Integration with Waste Treatment:** Using wastewater as a nutrient source for microalgae cultivation can provide a low-cost substrate while treating the wastewater, thus achieving dual benefits.
4. **Integration of Biohydrogen Production Into Microgrids:** Implementing biohydrogen production facilities within microgrids can significantly benefit rural and remote areas. By utilizing local biomass waste residues and cultivating microalgae biomass in wastewater treatment facilities, these regions can enhance their energy independence and sustainability. In an ideal scenario, microgrids powered by green hydrogen derived from biohydrogen may offer advantages over those using electrolytically produced hydrogen.

The potential for microalgae-based biofuels, including sustainable aviation fuel and biohydrogen, is significant, with the ability to transform the energy landscape and contribute to a sustainable and low-carbon future. Continued innovation and collaboration among research institutions, industry stakeholders, and policymakers are essential to realize the full potential of microalgae as a cornerstone of the biofuel industry.

Funding: This research was partially funded by the BMBF Project Professor “innengewinnung und Nachwuchsentwicklung zur Etablierung eines Centers of Advanced Scientific Education (CASE)” and partially supported by the Ministry of Science and Higher Education of Kazakhstan.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: No new data were created or analyzed in this study. Data sharing is not applicable to this article.

Conflicts of Interest: H.F. has been involved as a consultant in ENERTRAG SE.

References

1. Jabłońska-Trypuć, A.; Wołejko, E.; Ernazarovna, M.D.; Głowacka, A.; Sokołowska, G.; Wydro, U. Using Algae for Biofuel Production: A Review. *Energies* **2023**, *16*, 1758. [CrossRef]
2. Häder, D. On the Way to Mars—Flagellated Algae in Bioregenerative Life Support Systems Under Microgravity Conditions. *Front. Plant Sci.* **2020**, *10*, 1621. [CrossRef] [PubMed]
3. Ślusarczyk, J.; Adamska, E.; Czerwik-Marcinkowska, J. Fungi and Algae as Sources of Medicinal and Other Biologically Active Compounds: A Review. *Nutrients* **2021**, *13*, 3178. [CrossRef] [PubMed]
4. Henao, E.; Murphy, P.J.; Falfushynska, H.; Horyn, O.; Evans, D.M.; Klimaszcz, P.; Rzymiski, P. Polymethoxy-1-Alkenes Screening of *Chlorella* and *Spirulina* Food Supplements Coupled with In Vivo Toxicity Studies. *Toxins* **2020**, *12*, 111. [CrossRef] [PubMed]
5. Khan, M.I.; Shin, J.H.; Kim, J.D. The Promising Future of Microalgae: Current Status, Challenges, and Optimization of a Sustainable and Renewable Industry for Biofuels, Feed, and Other Products. *Microb. Cell Fact.* **2018**, *17*, 36. [CrossRef]
6. Chen, W.-H.; Lin, B.-J.; Huang, M.-Y.; Chang, J.-S. Thermochemical Conversion of Microalgal Biomass into Biofuels: A Review. *Bioresour. Technol.* **2015**, *184*, 314–327. [CrossRef]
7. Mahmood, T.; Hussain, N.; Shahbaz, A.; Mulla, S.I.; Iqbal, H.M.N.; Bilal, M. Sustainable Production of Biofuels from the Algae-Derived Biomass. *Bioprocess. Biosyst. Eng.* **2023**, *46*, 1077–1097. [CrossRef]
8. Kim, S.; Im, H.; Yu, J.; Kim, K.; Kim, M.; Lee, T. Biofuel Production from Euglena: Current Status and Techno-Economic Perspectives. *Bioresour. Technol.* **2023**, *371*, 128582. [CrossRef]
9. Mofijur, M.; Ashrafur Rahman, S.M.; Nguyen, L.N.; Mahlia, T.M.I.; Nghiem, L.D. Selection of Microalgae Strains for Sustainable Production of Aviation Biofuel. *Bioresour. Technol.* **2022**, *345*, 126408. [CrossRef]
10. Net Electricity Consumption Worldwide in Select Years from 1980 to 2022. Available online: <https://www.statista.com/statistics/280704/world-power-consumption> (accessed on 4 October 2024).
11. Shafiee, S.; Topal, E. When Will Fossil Fuel Reserves Be Diminished? *Energy Policy* **2009**, *37*, 181–189. [CrossRef]
12. Renewable Energy Targets. Available online: https://energy.ec.europa.eu/topics/renewable-energy/renewable-energy-directive-targets-and-rules/renewable-energy-targets_en (accessed on 4 October 2024).
13. Electricity 2030: Longterm Trends—Tasks for the Coming Years. Available online: https://www.bmwk.de/Redaktion/EN/Publikationen/electricity-2030-long-term-trends.pdf?__blob=publicationFile&v=1 (accessed on 4 October 2024).
14. Moiola, E.; Schildhauer, T. Negative CO₂ Emissions from Flexible Biofuel Synthesis: Concepts, Potentials, Technologies. *Renew. Sustain. Energy Rev.* **2022**, *158*, 112120. [CrossRef]
15. Falfushynska, H. Navigating Environmental Concerns: Assessing the Ecological Footprint of Photovoltaic-Produced Energy. *Environments* **2024**, *11*, 140. [CrossRef]
16. Lee, R.A.; Lavoie, J.-M. From First- to Third-Generation Biofuels: Challenges of Producing a Commodity from a Biomass of Increasing Complexity. *Anim. Front.* **2013**, *3*, 6–11. [CrossRef]
17. Yin, Z.; Zhu, L.; Li, S.; Hu, T.; Chu, R.; Mo, F.; Hu, D.; Liu, C.; Li, B. A Comprehensive Review on Cultivation and Harvesting of Microalgae for Biodiesel Production: Environmental Pollution Control and Future Directions. *Bioresour. Technol.* **2020**, *301*, 122804. [CrossRef]
18. Doliente, S.S.; Narayan, A.; Tapia, J.F.D.; Samsatli, N.J.; Zhao, Y.; Samsatli, S. Bio-Aviation Fuel: A Comprehensive Review and Analysis of the Supply Chain Components. *Front. Energy Res.* **2020**, *8*, 110. [CrossRef]
19. Islam, M.A.; Heimann, K.; Brown, R.J. Microalgae Biodiesel: Current Status and Future Needs for Engine Performance and Emissions. *Renew. Sustain. Energy Rev.* **2017**, *79*, 1160–1170. [CrossRef]
20. Wijffels, R.H.; Barbosa, M.J. An Outlook on Microalgal Biofuels. *Science* **2010**, *329*, 796–799. [CrossRef] [PubMed]
21. Tan, J.S.; Lee, S.Y.; Chew, K.W.; Lam, M.K.; Lim, J.W.; Ho, S.-H.; Show, P.L. A Review on Microalgae Cultivation and Harvesting, and Their Biomass Extraction Processing Using Ionic Liquids. *Bioengineered* **2020**, *11*, 116–129. [CrossRef]
22. The Future of the EU Algae Sector. Available online: [https://www.europarl.europa.eu/RegData/etudes/STUD/2023/733114/IPOL_STU\(2023\)733114_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/STUD/2023/733114/IPOL_STU(2023)733114_EN.pdf) (accessed on 4 October 2024).
23. Araújo, R.; Vázquez Calderón, F.; Sánchez López, J.; Azevedo, I.C.; Bruhn, A.; Fluch, S.; Garcia Tasende, M.; Ghaderiadekani, F.; Ilmjärvi, T.; Laurans, M.; et al. Current Status of the Algae Production Industry in Europe: An Emerging Sector of the Blue Bioeconomy. *Front. Mar. Sci.* **2021**, *7*, 626389. [CrossRef]
24. Abdur Razzak, S.; Bahar, K.; Islam, K.M.O.; Haniffa, A.K.; Faruque, M.O.; Hossain, S.M.Z.; Hossain, M.M. Microalgae Cultivation in Photobioreactors: Sustainable Solutions for a Greener Future. *Green. Chem. Eng.* **2024**, *5*, 418–439. [CrossRef]
25. Cui, X.; Yang, J.; Cui, M.; Zhang, W.; Zhao, J. Comparative Experiments of Two Novel Tubular Photobioreactors with an Inner Aerated Tube for Microalgal Cultivation: Enhanced Mass Transfer and Improved Biomass Yield. *Algal Res.* **2021**, *58*, 102364. [CrossRef]
26. Huang, Q.; Jiang, F.; Wang, L.; Yang, C. Design of Photobioreactors for Mass Cultivation of Photosynthetic Organisms. *Engineering* **2017**, *3*, 318–329. [CrossRef]
27. Standard Specification for Biodiesel Fuel Blend Stock (B100) for Middle Distillate Fuels. Available online: <https://www.astm.org/d6751-20a.html> (accessed on 4 October 2024).
28. EN 14214. Available online: https://www.chemeurope.com/en/encyclopedia/EN_14214.html (accessed on 4 October 2024).
29. Sharma, A.K.; Sharma, A.; Singh, Y.; Chen, W.-H. Production of a Sustainable Fuel from Microalgae *Chlorella minutissima* Grown in a 1500 L Open Raceway Ponds. *Biomass Bioenergy* **2021**, *149*, 106073. [CrossRef]

30. Schneider, R.D.C.D.S.; De Moura Lima, M.; Hoeltz, M.; De Farias Neves, F.; John, D.K.; De Azevedo, A. Life Cycle Assessment of Microalgae Production in a Raceway Pond with Alternative Culture Media. *Algal Res.* **2018**, *32*, 280–292. [\[CrossRef\]](#)
31. Rayen, F.; Behnam, T.; Dominique, P. Optimization of a Raceway Pond System for Wastewater Treatment: A Review. *Crit. Rev. Biotechnol.* **2019**, *39*, 422–435. [\[CrossRef\]](#)
32. Adesanya, V.O.; Cadena, E.; Scott, S.A.; Smith, A.G. Life Cycle Assessment on Microalgal Biodiesel Production Using a Hybrid Cultivation System. *Bioresour. Technol.* **2014**, *163*, 343–355. [\[CrossRef\]](#)
33. Yun, J.-H.; Cho, D.-H.; Lee, S.; Heo, J.; Tran, Q.-G.; Chang, Y.K.; Kim, H.-S. Hybrid Operation of Photobioreactor and Wastewater-Fed Open Raceway Ponds Enhances the Dominance of Target Algal Species and Algal Biomass Production. *Algal Res.* **2018**, *29*, 319–329. [\[CrossRef\]](#)
34. Kumar, M.S.; Hwang, J.-H.; Abou-Shanab, R.A.I.; Kabra, A.N.; Ji, M.-K.; Jeon, B.-H. Influence of CO₂ and Light Spectra on the Enhancement of Microalgal Growth and Lipid Content. *J. Renew. Sustain. Energy* **2014**, *6*, 063107. [\[CrossRef\]](#)
35. Assis, L.R.D.; Ferreira, J.; Assemany, P.P.; Teixeira, J.S.; Castro, J.D.S.; Pereira, H.A.; Calijuri, M.L. Environmental Benefits of a Hybrid System for Algal Biomass Production, Harvesting and Nutrient Recovery under a Life-Cycle Assessment. *Algal Res.* **2023**, *73*, 103163. [\[CrossRef\]](#)
36. Abreu, A.P.; Morais, R.C.; Teixeira, J.A.; Nunes, J. A Comparison between Microalgal Autotrophic Growth and Metabolite Accumulation with Heterotrophic, Mixotrophic and Photoheterotrophic Cultivation Modes. *Renew. Sustain. Energy Rev.* **2022**, *159*, 112247. [\[CrossRef\]](#)
37. Yun, H.-S.; Kim, Y.-S.; Yoon, H.-S. Effect of Different Cultivation Modes (Photoautotrophic, Mixotrophic, and Heterotrophic) on the Growth of *Chlorella* sp. and Biocompositions. *Front. Bioeng. Biotechnol.* **2021**, *9*, 774143. [\[CrossRef\]](#) [\[PubMed\]](#)
38. Proietti Tocca, G.; Agostino, V.; Menin, B.; Tommasi, T.; Fino, D.; Di Caprio, F. Mixotrophic and Heterotrophic Growth of Microalgae Using Acetate from Different Production Processes. *Rev. Environ. Sci. Biotechnol.* **2024**, *23*, 93–132. [\[CrossRef\]](#)
39. Grubišić, M.; Peremin, I.; Djedović, E.; Šantek, B.; Ivančić Šantek, M. Cultivation of a Novel Strain of *Chlorella vulgaris* S2 under Phototrophic, Mixotrophic, and Heterotrophic Conditions, and Effects on Biomass Growth and Composition. *Fermentation* **2024**, *10*, 270. [\[CrossRef\]](#)
40. Verma, R.; Kumari, K.V.L.K.; Srivastava, A.; Kumar, A. Photoautotrophic, Mixotrophic, and Heterotrophic Culture Media Optimization for Enhanced Microalgae Production. *J. Environ. Chem. Eng.* **2020**, *8*, 104149. [\[CrossRef\]](#)
41. Youssef, A.M.; Goma, M.; Mohamed, A.K.S.H.; El-Shanawany, A.-R.A. Enhancement of Biomass Productivity and Biochemical Composition of Alkaliphilic Microalgae by Mixotrophic Cultivation Using Cheese Whey for Biofuel Production. *Environ. Sci. Pollut. Res.* **2024**, *31*, 42875–42888. [\[CrossRef\]](#)
42. Zuliani, L.; Frison, N.; Jelic, A.; Fatone, F.; Bolzonella, D.; Ballottari, M. Microalgae Cultivation on Anaerobic Digestate of Municipal Wastewater, Sewage Sludge and Agro-Waste. *Int. J. Mol. Sci.* **2016**, *17*, 1692. [\[CrossRef\]](#)
43. Kong, W.; Kong, J.; Feng, S.; Yang, T.; Xu, L.; Shen, B.; Bi, Y.; Lyu, H. Cultivation of Microalgae–Bacteria Consortium by Waste Gas–Waste Water to Achieve CO₂ Fixation, Wastewater Purification and Bioproducts Production. *Biotechnol. Biofuels* **2024**, *17*, 26. [\[CrossRef\]](#)
44. Velásquez-Orta, S.B.; Yáñez-Noguez, I.; Ramírez, I.M.; Ledesma, M.T.O. Pilot-Scale Microalgae Cultivation and Wastewater Treatment Using High-Rate Ponds: A Meta-Analysis. *Environ. Sci. Pollut. Res.* **2024**, *31*, 46994–47021. [\[CrossRef\]](#)
45. Zewdie, D.T.; Ali, A.Y. Cultivation of Microalgae for Biofuel Production: Coupling with Sugarcane-Processing Factories. *Energy Sustain. Soc.* **2020**, *10*, 27. [\[CrossRef\]](#)
46. Osman, M.E.H.; Abo-Shady, A.M.; Gheda, S.F.; Desoki, S.M.; Elshobary, M.E. Unlocking the Potential of Microalgae Cultivated on Wastewater Combined with Salinity Stress to Improve Biodiesel Production. *Environ. Sci. Pollut. Res.* **2023**, *30*, 114610–114624. [\[CrossRef\]](#)
47. Koley, S.; Prasad, S.; Bagchi, S.K.; Mallick, N. Development of a Harvesting Technique for Large-Scale Microalgal Harvesting for Biodiesel Production. *RSC Adv.* **2017**, *7*, 7227–7237. [\[CrossRef\]](#)
48. Wrede, D.; Taha, M.; Miranda, A.F.; Kadali, K.; Stevenson, T.; Ball, A.S.; Mouradov, A. Co-Cultivation of Fungal and Microalgal Cells as an Efficient System for Harvesting Microalgal Cells, Lipid Production and Wastewater Treatment. *PLoS ONE* **2014**, *9*, e113497. [\[CrossRef\]](#) [\[PubMed\]](#)
49. Najjar, Y.S.H.; Abu-Shamleh, A. Harvesting of Microalgae by Centrifugation for Biodiesel Production: A Review. *Algal Res.* **2020**, *51*, 102046. [\[CrossRef\]](#)
50. Bwapwa, J.K.; Anandraj, A.; Trois, C. Possibilities for Conversion of Microalgae Oil into Aviation Fuel: A Review. *Renew. Sustain. Energy Rev.* **2017**, *80*, 1345–1354. [\[CrossRef\]](#)
51. Luo, S.; Wu, X.; Jiang, H.; Yu, M.; Liu, Y.; Min, A.; Li, W.; Ruan, R. Edible Fungi-Assisted Harvesting System for Efficient Microalgae Bio-Flocculation. *Bioresour. Technol.* **2019**, *282*, 325–330. [\[CrossRef\]](#)
52. Mata, T.M.; Martins, A.A.; Caetano, N.S. Microalgae for Biodiesel Production and Other Applications: A Review. *Renew. Sustain. Energy Rev.* **2010**, *14*, 217–232. [\[CrossRef\]](#)
53. De Morais, E.G.; Sampaio, I.C.F.; Gonzalez-Flo, E.; Ferrer, I.; Uggetti, E.; García, J. Microalgae Harvesting for Wastewater Treatment and Resources Recovery: A Review. *New Biotechnol.* **2023**, *78*, 84–94. [\[CrossRef\]](#)
54. Zhu, L.; Li, Z.; Hiltunen, E. Microalgae *Chlorella vulgaris* Biomass Harvesting by Natural Flocculant: Effects on Biomass Sedimentation, Spent Medium Recycling and Lipid Extraction. *Biotechnol. Biofuels* **2018**, *11*, 183. [\[CrossRef\]](#)

55. Milledge, J.J.; Heaven, S. A Review of the Harvesting of Micro-Algae for Biofuel Production. *Rev. Environ. Sci. Biotechnol.* **2013**, *12*, 165–178. [\[CrossRef\]](#)
56. Khoo, K.S.; Ahmad, I.; Chew, K.W.; Iwamoto, K.; Bhatnagar, A.; Show, P.L. Enhanced Microalgal Lipid Production for Biofuel Using Different Strategies Including Genetic Modification of Microalgae: A Review. *Prog. Energy Combust. Sci.* **2023**, *96*, 101071. [\[CrossRef\]](#)
57. Sun, X.-M.; Ren, L.-J.; Zhao, Q.-Y.; Ji, X.-J.; Huang, H. Microalgae for the Production of Lipid and Carotenoids: A Review with Focus on Stress Regulation and Adaptation. *Biotechnol. Biofuels* **2018**, *11*, 272. [\[CrossRef\]](#) [\[PubMed\]](#)
58. Deshmukh, S.; Bala, K.; Kumar, R. Selection of Microalgae Species Based on Their Lipid Content, Fatty Acid Profile and Apparent Fuel Properties for Biodiesel Production. *Environ. Sci. Pollut. Res.* **2019**, *26*, 24462–24473. [\[CrossRef\]](#)
59. Morales, M.; Aflalo, C.; Bernard, O. Microalgal Lipids: A Review of Lipids Potential and Quantification for 95 Phytoplankton Species. *Biomass Bioenergy* **2021**, *150*, 106108. [\[CrossRef\]](#)
60. Demirbas, A.; Fatih Demirbas, M. Importance of Algae Oil as a Source of Biodiesel. *Energy Convers. Manag.* **2011**, *52*, 163–170. [\[CrossRef\]](#)
61. Mostafa, S.S.M.; El-Gendy, N.S. Evaluation of Fuel Properties for Microalgae *Spirulina Platensis* Bio-Diesel and Its Blends with Egyptian Petro-Diesel. *Arab. J. Chem.* **2017**, *10*, S2040–S2050. [\[CrossRef\]](#)
62. Lee, S.-J.; Go, S.; Jeong, G.-T.; Kim, S.-K. Oil Production from Five Marine Microalgae for the Production of Biodiesel. *Biotechnol. Bioprocess Eng.* **2011**, *16*, 561–566. [\[CrossRef\]](#)
63. Pugliese, A.; Biondi, L.; Bartocci, P.; Fantozzi, F. *Selenastrum Capricornutum* a New Strain of Algae for Biodiesel Production. *Fermentation* **2020**, *6*, 46. [\[CrossRef\]](#)
64. Hawrot-Paw, M.; Ratowski, P.; Koniuszy, A.; Golimowski, W.; Teleszko, M.; Grygier, A. Fatty Acid Profile of Microalgal Oils as a Criterion for Selection of the Best Feedstock for Biodiesel Production. *Energies* **2021**, *14*, 7334. [\[CrossRef\]](#)
65. Andrew, A.R.; Yong, W.T.L.; Misson, M.; Anton, A.; Chin, G.J.W.L. Selection of Tropical Microalgae Species for Mass Production Based on Lipid and Fatty Acid Profiles. *Front. Energy Res.* **2022**, *10*, 912904. [\[CrossRef\]](#)
66. Yang, Y.; Du, L.; Hosokawa, M.; Miyashita, K. Total Lipids Content, Lipid Class and Fatty Acid Composition of Ten Species of Microalgae. *J. Oleo Sci.* **2020**, *69*, 1181–1189. [\[CrossRef\]](#)
67. Enwereuzoh, U.; Harding, K.; Low, M. Characterization of Biodiesel Produced from Microalgae Grown on Fish Farm Wastewater. *SN Appl. Sci.* **2020**, *2*, 970. [\[CrossRef\]](#)
68. Rodríguez-Palacio, M.C.; Cabrera-Cruz, R.B.E.; Rolón-Aguilar, J.C.; Tobías-Jaramillo, R.; Martínez-Hernández, M.; Lozano-Ramírez, C. The Cultivation of Five Microalgae Species and Their Potential for Biodiesel Production. *Energy Sustain. Soc.* **2022**, *12*, 10. [\[CrossRef\]](#)
69. Anandraj, A.; White, S.; Bwapwa, J.K. Temperature and Nutrient Coupled Stress on Microalgal Neutral Lipids for Low-carbon Fuels Production. *Biofuels Bioprod. Bioref.* **2021**, *15*, 1073–1086. [\[CrossRef\]](#)
70. Gui, J.; Chen, S.; Luo, G.; Wu, Z.; Fan, Y.; Yao, L.; Xu, H. Nutrient Deficiency and an Algicidal Bacterium Improved the Lipid Profiles of a Novel Promising Oleaginous Dinoflagellate, *Prorocentrum donghaiense*, for Biodiesel Production. *Appl. Environ. Microbiol.* **2021**, *87*, e01159–21. [\[CrossRef\]](#) [\[PubMed\]](#)
71. El-Sheekh, M.M.; Mansour, H.M.; Bedaiwy, M.Y.; El-shenody, R.A. Influence of Nutrient Supplementation and Stress Conditions on the Biomass and Lipid Production of *Microchloropsis Salina* for Biodiesel Production. *Biomass Conv. Bioref.* **2022**. [\[CrossRef\]](#)
72. Udayan, A.; Pandey, A.K.; Sirohi, R.; Sreekumar, N.; Sang, B.-I.; Sim, S.J.; Kim, S.H.; Pandey, A. Production of Microalgae with High Lipid Content and Their Potential as Sources of Nutraceuticals. *Phytochem. Rev.* **2023**, *22*, 833–860. [\[CrossRef\]](#)
73. Kim, S.; Kim, M.; Chang, Y.K.; Kim, D. Lipid Production under a Nutrient-Sufficient Condition Outperforms Starvation Conditions Due to a Natural Polarization of Lipid Content in Algal Biofilm. *Fuel* **2023**, *339*, 126902. [\[CrossRef\]](#)
74. Ratowski, P.; Hawrot-Paw, M. Influence of Nutrient-Stress Conditions on *Chlorella vulgaris* Biomass Production and Lipid Content. *Catalysts* **2021**, *11*, 573. [\[CrossRef\]](#)
75. Abd El Baky, H.; El Baroty, G. Cultivation of *Pseudochlorella Pringsheimii* for Biodiesel Production in a Scalable Indoor Photobioreactor: Case Studies from Egypt. *J. Genet. Eng. Biotechnol.* **2023**, *21*, 25. [\[CrossRef\]](#)
76. El-Sheekh, M.M.; Galal, H.R.; Mousa, A.S.H.; Farghl, A.A.M. Impact of Macronutrients and Salinity Stress on Biomass and Biochemical Constituents in *Monoraphidium Braunii* to Enhance Biodiesel Production. *Sci. Rep.* **2024**, *14*, 2725. [\[CrossRef\]](#)
77. Yun, C.-J.; Hwang, K.-O.; Han, S.-S.; Ri, H.-G. The Effect of Salinity Stress on the Biofuel Production Potential of Freshwater Microalgae *Chlorella vulgaris* YH703. *Biomass Bioenergy* **2019**, *127*, 105277. [\[CrossRef\]](#)
78. Vrana, I.; Bakija Alempijević, S.; Novosel, N.; Ivošević DeNardis, N.; Žigon, D.; Ogrinc, N.; Gašparović, B. Hyposalinity Induces Significant Polar Lipid Remodeling in the Marine Microalga *Dunaliella tertiolecta* (Chlorophyceae). *J. Appl. Phycol.* **2022**, *34*, 1457–1470. [\[CrossRef\]](#)
79. Khatoon, H.; Haris, N.; Banerjee, S.; Rahman, N.A.; Begum, H.; Mian, S.; Abol-Munafi, A.B.; Endut, A. Effects of Different Salinities on the Growth and Proximate Composition of *Dunaliella* sp. Isolated from South China Sea at Different Growth Phases. *Process Saf. Environ. Prot.* **2017**, *112*, 280–287. [\[CrossRef\]](#)
80. Conde, T.; Aveiro, S.; Melo, T.; Santos, T.; Neves, B.; Domingues, P.; Varela, J.; Pereira, H.; Domingues, M.R. Cross-Stress Lipid Response of *Tetraselmis striata* CTP4 to Temperature and Salinity Variation. *Algal Res.* **2023**, *74*, 103218. [\[CrossRef\]](#)
81. Singh, R.P.; Yadav, P.; Kumar, A.; Hashem, A.; Avila-Quezada, G.D.; Abd_Allah, E.F.; Gupta, R.K. Salinity-Induced Physiochemical Alterations to Enhance Lipid Content in Oleaginous Microalgae *Scenedesmus* sp. BHU1 via Two-Stage Cultivation for Biodiesel Feedstock. *Microorganisms* **2023**, *11*, 2064. [\[CrossRef\]](#)

82. Yang, Z.-Y.; Huang, K.-X.; Zhang, Y.-R.; Yang, L.; Zhou, J.-L.; Yang, Q.; Gao, F. Efficient Microalgal Lipid Production Driven by Salt Stress and Phytohormones Synergistically. *Bioresour. Technol.* **2023**, *367*, 128270. [CrossRef]
83. Li, H.; Sun, X.; Sun, Y.; Ye, L.; Xue, H.; Gao, F.; Yang, Y. Enhancing Lipid Production in *Chlorella* under Successive Stresses of Periodic Micro-Current and Salinity: Performance and Contribution. *Chem. Eng. J.* **2024**, *486*, 150409. [CrossRef]
84. Chen, Y.; Xu, C.; Vaidyanathan, S. Influence of Gas Management on Biochemical Conversion of CO₂ by Microalgae for Biofuel Production. *Appl. Energy* **2020**, *261*, 114420. [CrossRef]
85. Wu, S.; Gu, W.; Huang, A.; Li, Y.; Kumar, M.; Lim, P.E.; Huan, L.; Gao, S.; Wang, G. Elevated CO₂ Improves Both Lipid Accumulation and Growth Rate in the Glucose-6-Phosphate Dehydrogenase Engineered *Phaeodactylum Tricornutum*. *Microb. Cell Fact.* **2019**, *18*, 161. [CrossRef]
86. Widjaja, A.; Chien, C.-C.; Ju, Y.-H. Study of Increasing Lipid Production from Fresh Water Microalgae *Chlorella vulgaris*. *J. Taiwan Inst. Chem. Eng.* **2009**, *40*, 13–20. [CrossRef]
87. Levitan, O.; Dinamarca, J.; Zelzion, E.; Gorbunov, M.Y.; Falkowski, P.G. An RNA Interference Knock-down of Nitrate Reductase Enhances Lipid Biosynthesis in the Diatom *Phaeodactylum tricornutum*. *Plant J.* **2015**, *84*, 963–973. [CrossRef]
88. Takahashi, K.; Ide, Y.; Hayakawa, J.; Yoshimitsu, Y.; Fukuhara, I.; Abe, J.; Kasai, Y.; Harayama, S. Lipid Productivity in TALEN-Induced Starchless Mutants of the Unicellular Green Alga *Coccomyxa* sp. Strain Obi. *Algal Res.* **2018**, *32*, 300–307. [CrossRef]
89. Kurita, T.; Moroi, K.; Iwai, M.; Okazaki, K.; Shimizu, S.; Nomura, S.; Saito, F.; Maeda, S.; Takami, A.; Sakamoto, A.; et al. Efficient and Multiplexable Genome Editing Using Platinum TALENs in Oleaginous Microalga, *Nannochloropsis oceanica* NIES-2145. *Genes Cells* **2020**, *25*, 695–702. [CrossRef]
90. Dhokane, D.; Shaikh, A.; Yadav, A.; Giri, N.; Bandyopadhyay, A.; Dasgupta, S.; Bhadra, B. CRISPR-Based Bioengineering in Microalgae for Production of Industrially Important Biomolecules. *Front. Bioeng. Biotechnol.* **2023**, *11*, 1267826. [CrossRef]
91. Lin, W.-R.; Ng, I.-S. Development of CRISPR/Cas9 System in *Chlorella vulgaris* FSP-E to Enhance Lipid Accumulation. *Enzym. Microb. Technol.* **2020**, *133*, 109458. [CrossRef]
92. Kasai, Y.; Takagi, S.; Ota, S.; Ishii, K.; Takeshita, T.; Kawano, S.; Harayama, S. Development of a CRISPR/Cas9-Mediated Gene-Editing Method to Isolate a Mutant of the Unicellular Green Alga *Parachlorella Kessleri* Strain NIES-2152 with Improved Lipid Productivity. *Biotechnol. Biofuels* **2024**, *17*, 36. [CrossRef]
93. Chang, K.S.; Kim, J.; Park, H.; Hong, S.-J.; Lee, C.-G.; Jin, E. Enhanced Lipid Productivity in AGP Knockout Marine Microalga *Tetraselmis* sp. Using a DNA-Free CRISPR-Cas9 RNP Method. *Bioresour. Technol.* **2020**, *303*, 122932. [CrossRef]
94. ASTM Biodiesel Specifications. Available online: <https://afdc.energy.gov/fuels/biodiesel-specifications> (accessed on 4 October 2024).
95. ASTM D975-20a; Specification for Diesel Fuel. ASTM International: West Conshohocken, PA, USA, 2019.
96. Das, S.; Chowdhury, A. An Exploration of Biodiesel for Application in Aviation and Automobile Sector. *Energy Nexus* **2023**, *10*, 100204. [CrossRef]
97. Alptekin, E.; Canakci, M. Determination of the Density and the Viscosities of Biodiesel–Diesel Fuel Blends. *Renew. Energy* **2008**, *33*, 2623–2630. [CrossRef]
98. Bharti, R.K.; Kaushal, C.; Singh, A.; Dhar, D.W.; Babu, R.; Kaushik, A. Evaluation of Fuel Properties for Possible Biodiesel Output Based on the Fatty Acid Composition of Oleaginous Plants and Microalgae. *Sci. Total Environ.* **2024**, *918*, 170448. [CrossRef] [PubMed]
99. Mishra, S.; Bukkarapu, K.R.; Krishnasamy, A. A Composition Based Approach to Predict Density, Viscosity and Surface Tension of Biodiesel Fuels. *Fuel* **2021**, *285*, 119056. [CrossRef]
100. Park, S.H.; Khan, N.; Lee, S.; Zimmermann, K.; DeRosa, M.; Hamilton, L.; Hudson, W.; Hyder, S.; Serratos, M.; Sheffield, E.; et al. Biodiesel Production from Locally Sourced Restaurant Waste Cooking Oil and Grease: Synthesis, Characterization, and Performance Evaluation. *ACS Omega* **2019**, *4*, 7775–7784. [CrossRef]
101. Bessi res, D.; Bazile, J.-P.; Tanh, X.N.T.; Garc a-Cuadra, F.; Acien, F.G. Thermophysical Behavior of Three Algal Biodiesels over Wide Ranges of Pressure and Temperature. *Fuel* **2018**, *233*, 497–503. [CrossRef]
102. Bucy, H.B.; Baumgardner, M.E.; Marchese, A.J. Chemical and Physical Properties of Algal Methyl Ester Biodiesel Containing Varying Levels of Methyl Eicosapentaenoate and Methyl Docosahexaenoate. *Algal Res.* **2012**, *1*, 57–69. [CrossRef]
103. Al-Ansari, M.M.; Al-Humaid, L.; Al-Dahmash, N.D.; Aldawsari, M. Assessing the Benefits of *Chlorella vulgaris* Microalgal Biodiesel for Internal Combustion Engines: Energy and Exergy Analyses. *Fuel* **2023**, *344*, 128055. [CrossRef]
104. Sharma, A.K.; Sahoo, P.K.; Singhal, S. Feasibility of Biodiesel Production from *Chlorella vulgaris* Grown in a Flat Plate Photobioreactor under Outdoor Conditions. *Int. J. ChemTech Res.* **2015**, *8*, 671–678.
105. Karmakar, R.; Kundu, K.; Rajor, A. Fuel Properties and Emission Characteristics of Biodiesel Produced from Unused Algae Grown in India. *Pet. Sci.* **2018**, *15*, 385–395. [CrossRef]
106. Ejim, C.E.; Fleck, B.A.; Amirfazli, A. Analytical Study for Atomization of Biodiesels and Their Blends in a Typical Injector: Surface Tension and Viscosity Effects. *Fuel* **2007**, *86*, 1534–1544. [CrossRef]
107. Das, M.; Sarkar, M.; Datta, A.; Santra, A.K. Study on Viscosity and Surface Tension Properties of Biodiesel–Diesel Blends and Their Effects on Spray Parameters for CI Engines. *Fuel* **2018**, *220*, 769–779. [CrossRef]
108. Al-Iwayzy, S.; Yusaf, T.; Al-Juboori, R. Biofuels from the Fresh Water Microalgae *Chlorella vulgaris* (FWM-CV) for Diesel Engines. *Energies* **2014**, *7*, 1829–1851. [CrossRef]
109. Pillet, F.; Dague, E.; Pe ar Ili , J.; Ru i , I.; Rols, M.-P.; Ivo evi  DeNardis, N. Changes in Nanomechanical Properties and Adhesion Dynamics of Algal Cells during Their Growth. *Bioelectrochemistry* **2019**, *127*, 154–162. [CrossRef] [PubMed]

110. Lai, Y.S.; Francesco, F.D.; Aguinaga, A.; Parameswaran, P.; Rittmann, B.E. Improving Lipid Recovery from *Scenedesmus* Wet Biomass by Surfactant-Assisted Disruption. *Green Chem.* **2016**, *18*, 1319–1326. [CrossRef]
111. Shimi, E.; Sheltawy, E.M.; Diwani, E. Reactive Extraction of Microalgae for Biodiesel Production; an Optimization Study. *Environ. Sci. Eng. Chem.* **2014**. Available online: <https://api.semanticscholar.org/CorpusID:52467259> (accessed on 4 October 2024).
112. Salam, K.A. Reactive Extraction of Microalgae for Biodiesel Production. Ph.D. Thesis, Newcastle University, Newcastle upon Tyne, UK, 2015. Available online: <https://core.ac.uk/download/pdf/153779776.pdf> (accessed on 4 October 2024).
113. Salam, K.A.; Velasquez-Orta, S.B.; Harvey, A.P. Surfactant-Assisted Direct Biodiesel Production from Wet *Nannochloropsis oculata* by in Situ Transesterification/Reactive Extraction. *Biofuel Res. J.* **2016**, *3*, 366–371. [CrossRef]
114. Racheva, R.; Tietgens, N.; Kerner, M.; Smirnova, I. In Situ Continuous Countercurrent Cloud Point Extraction of Microalgae Cultures. *Sep. Purif. Technol.* **2018**, *190*, 268–277. [CrossRef]
115. Lim, J.H.K.; Gan, Y.Y.; Ong, H.C.; Lau, B.F.; Chen, W.-H.; Chong, C.T.; Ling, T.C.; Klemeš, J.J. Utilization of Microalgae for Bio-Jet Fuel Production in the Aviation Sector: Challenges and Perspective. *Renew. Sustain. Energy Rev.* **2021**, *149*, 111396. [CrossRef]
116. Atnoorkar, S.; Wiatrowski, M.; Newes, E.; Davis, R.; Peterson, S. Algae to HEFA: Economics and Potential Deployment in the United States. *Biofuels Bioprod. Bioref.* **2024**, *18*, 1121–1136. [CrossRef]
117. Algal Biofuels: Long-Term Energy Benefits Drive U.S. Research. Available online: <https://www.energy.gov/eere/bioenergy/articles/algal-biofuels-long-term-energy-benefits-drive-us-research> (accessed on 4 October 2024).
118. Gasoline and Diesel Fuel Update. Available online: <https://www.eia.gov/petroleum/gasdiesel> (accessed on 4 October 2024).
119. Sun, J.; Xiong, X.; Wang, M.; Du, H.; Li, J.; Zhou, D.; Zuo, J. Microalgae Biodiesel Production in China: A Preliminary Economic Analysis. *Renew. Sustain. Energy Rev.* **2019**, *104*, 296–306. [CrossRef]
120. Yang, Y.; Zhang, B.; Cheng, J.; Pu, S. Socio-Economic Impacts of Algae-Derived Biodiesel Industrial Development in China: An Input–Output Analysis. *Algal Res.* **2015**, *9*, 74–81. [CrossRef]
121. Germans and French Bring Algal Biofuel to Indian Ocean. Available online: <https://advancedbiofuelsusa.info/germans-and-french-bring-algal-biofuel-to-indian-ocean> (accessed on 4 October 2024).
122. Bavarian Facility Can Now Grow Global Strains of Algae for Aerospace Biofuel Production. Available online: <https://www.labiotech.eu/trends-news/aviation-biofuels-bavarian-algae-plant> (accessed on 4 October 2024).
123. New EU-Funded FUELGAE Project Aims to Resurrect Algae as a Promising Source for Aviation Biofuels. 2024. Available online: <https://www.greenairnews.com/?p=5193> (accessed on 4 October 2024).
124. D-FACTORY: The Micro-Algae Biorefinery. Available online: <https://www.ifeu.de/en/project/d-factory-the-micro-algae-biorefinery> (accessed on 4 October 2024).
125. Raheem, A.; Abbasi, S.A.; Mangi, F.H.; Ahmed, S.; He, Q.; Ding, L.; Memon, A.A.; Zhao, M.; Yu, G. Gasification of Algal Residue for Synthesis Gas Production. *Algal Res.* **2021**, *58*, 102411. [CrossRef]
126. Al-Rabiah, A.A.; Al-Dawsari, J.N.; Ajbar, A.M.; Al Darwish, R.K.; Abdelaziz, O.Y. Development of a Biomass Gasification Process for the Coproduction of Methanol and Power from Red Sea Microalgae. *Energies* **2022**, *15*, 7890. [CrossRef]
127. Ebadi, A.G.; Hisoriev, H.; Zarnegar, M.; Ahmadi, H. Hydrogen and Syngas Production by Catalytic Gasification of Algal Biomass (*Cladophora glomerata* L.) Using Alkali and Alkaline-Earth Metals Compounds. *Environ. Technol.* **2019**, *40*, 1178–1184. [CrossRef] [PubMed]
128. Raheem, A.; Dupont, V.; Channa, A.Q.; Zhao, X.; Vuppaladadiyam, A.K.; Taufiq-Yap, Y.-H.; Zhao, M.; Harun, R. Parametric Characterization of Air Gasification of *Chlorella vulgaris* Biomass. *Energy Fuels* **2017**, *31*, 2959–2969. [CrossRef]
129. Marangon, B.B.; Castro, J.D.S.; Assemany, P.P.; Machado, N.A.; Calijuri, M.L. Wastewater-Grown Microalgae Biomass as a Source of Sustainable Aviation Fuel: Life Cycle Assessment Comparing Hydrothermal Routes. *J. Environ. Manag.* **2024**, *360*, 121164. [CrossRef]
130. Kenkel, P.; Wassermann, T.; Zondervan, E. Renewable Fuels from Integrated Power- and Biomass-to-X Processes: A Superstructure Optimization Study. *Processes* **2022**, *10*, 1298. [CrossRef]
131. Yousefian, F.; Babatabar, M.A.; Eshaghi, M.; Poor, S.M.; Tavasoli, A. Pyrolysis of Rice Husk, Coconut Shell, and *Cladophora Glomerata* Algae and Application of the Produced Biochars as Support for Cobalt Catalyst in Fischer–Tropsch Synthesis. *Fuel Process. Technol.* **2023**, *247*, 107818. [CrossRef]
132. Bayat, F.; Pirbazari, S.M.; Shojaei, N.; Kiani, S.; Tavasoli, A. Green Catalyst Innovation: Enhanced Fischer–Tropsch Synthesis Using Potassium-Promoted Cobalt Catalysts Supported on Pyrolyzed Peanut Shells and *Cladophora Glomerata* Modified Biochars. *Fuel Process. Technol.* **2024**, *258*, 108094. [CrossRef]
133. Chiamonti, D.; Talluri, G.; Scarlat, N.; Prussi, M. The Challenge of Forecasting the Role of Biofuel in EU Transport Decarbonisation at 2050: A Meta-Analysis Review of Published Scenarios. *Renew. Sustain. Energy Rev.* **2021**, *139*, 110715. [CrossRef]
134. Sharma, V.; Hossain, A.K.; Duraisamy, G.; Griffiths, G. Microalgal Biodiesel: A Challenging Route toward a Sustainable Aviation Fuel. *Fermentation* **2023**, *9*, 907. [CrossRef]
135. Velmozhina, K.; Shinkevich, P.; Zhazhkov, V.; Politaeva, N.; Korablev, V.; Vladimirov, I.; Morales, T.C. Production of Biohydrogen from Microalgae Biomass after Wastewater Treatment and Air Purification from CO₂. *Processes* **2023**, *11*, 2978. [CrossRef]
136. Falfushynska, H.; Zhadan, V.; Holz, M. Advancement and Assessment of Power-to-X Strategies as a Significant Contribution for the De-Fossilization of Economies. Universitäts- und Landesbibliothek Sachsen-Anhalt. *Proc. Int. Conf. Appl. Innov. IT* **2024**, *12*, 233–241. [CrossRef]
137. Zhang, J.; Xue, D.; Wang, C.; Fang, D.; Cao, L.; Gong, C. Genetic Engineering for Biohydrogen Production from Microalgae. *iScience* **2023**, *26*, 107255. [CrossRef] [PubMed]

138. Huang, Z.; Zhang, J.; Pan, M.; Hao, Y.; Hu, R.; Xiao, W.; Li, G.; Lyu, T. Valorisation of Microalgae Residues after Lipid Extraction: Pyrolysis Characteristics for Biofuel Production. *Biochem. Eng. J.* **2022**, *179*, 108330. [CrossRef]
139. Wang, K.; Khoo, K.S.; Chew, K.W.; Selvarajoo, A.; Chen, W.-H.; Chang, J.-S.; Show, P.L. Microalgae: The Future Supply House of Biohydrogen and Biogas. *Front. Energy Res.* **2021**, *9*, 660399. [CrossRef]
140. Falfushynska, H.; Holz, M. Power-to-X Strategies: A Key Driver for Decarbonization and Renewable Energy Integration in Economies. Lecture Notes in Networks and Systems. *Lect. Notes Netw. Syst.* **2024**, in press.
141. Chen, S.; Qu, D.; Xiao, X.; Miao, X. Biohydrogen Production with Lipid-Extracted *Dunaliella* Biomass and a New Strain of Hyper-Thermophilic Archaeon *Thermococcus eurythermalis* A501. *Int. J. Hydrogen Energy* **2020**, *45*, 12721–12730. [CrossRef]
142. Wang, J.; Yin, Y. Fermentative Hydrogen Production Using Pretreated Microalgal Biomass as Feedstock. *Microb. Cell Fact.* **2018**, *17*, 22. [CrossRef]
143. Mustafi, N.N.; Hossain, M.I.; Ahammad, M.F.; Naz, S. Biohydrogen Production from *Euglena acus* Microalgae Available in Bangladesh. *MethodsX* **2023**, *10*, 101976. [CrossRef]
144. Biohydrogen: Affordable, Green and yet Overlooked. 2023. Available online: <https://www.europeanbiogas.eu/biohydrogen-affordable-green-and-yet-overlooked> (accessed on 4 October 2024).
145. Jiao, H.; Tsigkou, K.; Elsamahy, T.; Pispas, K.; Sun, J.; Manthos, G.; Schagerl, M.; Sventzouri, E.; Al-Tohamy, R.; Kornaros, M.; et al. Recent Advances in Sustainable Hydrogen Production from Microalgae: Mechanisms, Challenges, and Future Perspectives. *Ecotoxicol. Environ. Saf.* **2024**, *270*, 115908. [CrossRef]
146. Genç, Ş.; Koku, H. A Preliminary Techno-Economic Analysis of Photofermentative Hydrogen Production. *Int. J. Hydrogen Energy* **2024**, *52*, 212–222. [CrossRef]
147. Somers, M.D.; Chen, P.; Clippinger, J.; Cruce, J.R.; Davis, R.; Lammers, P.J.; Quinn, J.C. Techno-Economic and Life-Cycle Assessment of Fuel Production from Mixotrophic *Galdieria Sulphuraria* Microalgae on Hydrolysate. *Algal Res.* **2021**, *59*, 102419. [CrossRef]
148. Yang, D.-W.; Syn, J.-W.; Hsieh, C.-H.; Huang, C.-C.; Chien, L.-F. Genetically Engineered Hydrogenases Promote Biophotocatalysis-Mediated H₂ Production in the Green Alga *Chlorella* sp. DT. *Int. J. Hydrogen Energy* **2019**, *44*, 2533–2545. [CrossRef]
149. Min Woon, J.; Shiong Khoo, K.; Akermi, M.; Alanazi, M.M.; Wei Lim, J.; Jing Chan, Y.; Sean Goh, P.; Silas Chidi, B.; Kee Lam, M.; Zaini, J.; et al. Reviewing Biohydrogen Production from Microalgal Cells through Fundamental Mechanisms, Enzymes and Factors That Engendering New Challenges and Prospects. *Fuel* **2023**, *346*, 128312. [CrossRef]
150. Babu Pallam, R.; Ramamoorthy, N.K.; Rakshit, S.; Jaiswal, K.K.; Sarma, V.V. Microalgae for the Sustainable Production of Biohydrogen. In *Clean Energy Transition-via-Biomass Resource Utilization*; Kumar, S., Sundaramurthy, S., Kumar, D., Chandel, A.K., Eds.; Green Energy and Technology; Springer Nature: Singapore, 2024; pp. 151–175. [CrossRef]
151. Bhatia, S.K.; Jagtap, S.S.; Bedekar, A.A.; Bhatia, R.K.; Rajendran, K.; Pugazhendhi, A.; Rao, C.V.; Atabani, A.E.; Kumar, G.; Yang, Y.-H. Renewable Biohydrogen Production from Lignocellulosic Biomass Using Fermentation and Integration of Systems with Other Energy Generation Technologies. *Sci. Total Environ.* **2021**, *765*, 144429. [CrossRef] [PubMed]
152. Xia, A.; Cheng, J.; Song, W.; Su, H.; Ding, L.; Lin, R.; Lu, H.; Liu, J.; Zhou, J.; Cen, K. Fermentative Hydrogen Production Using Algal Biomass as Feedstock. *Renew. Sustain. Energy Rev.* **2015**, *51*, 209–230. [CrossRef]
153. Kiehadrouinezhad, M.; Merabet, A.; Ghenai, C.; Abo-Khalil, A.G.; Salameh, T. The Role of Biofuels for Sustainable MicrogridsF: A Path towards Carbon Neutrality and the Green Economy. *Heliyon* **2023**, *9*, e13407. [CrossRef]
154. Navas, S.J.; Cabello González, G.M.; Pino, A.; Pino, F.J. Integration of Microgrids in Chemical Industries with Hydrogen as a Byproduct: Styrene Production Case Study. *Int. J. Hydrogen Energy* **2024**, *59*, 947–957. [CrossRef]
155. Eloffy, M.G.; Elgarahy, A.M.; Saber, A.N.; Hammad, A.; El-Sherif, D.M.; Shehata, M.; Mohsen, A.; Elwakeel, K.Z. Biomass-to-Sustainable Biohydrogen: Insights into the Production Routes, and Technical Challenges. *Chem. Eng. J. Adv.* **2022**, *12*, 100410. [CrossRef]
156. Mandalika, A.S.; Chou, K.J.; Decker, S.R. Biohydrogen: Prospects for Industrial Utilization and Energy Resiliency in Rural Communities. *Front. Ind. Microbiol.* **2024**, *2*, 1428686. [CrossRef]
157. Biofuels Saved 9.7 Million Tonnes of CO₂ Equivalent in 2019. Available online: https://www.ble.de/SharedDocs/Meldungen/EN/2021/210226_biofuels.html (accessed on 4 October 2024).
158. Final Report Summary-BIOALGAESORB. Available online: <https://cordis.europa.eu/project/id/243752/reporting/it> (accessed on 4 October 2024).
159. Biofuels Dashboard 2023. Available online: <https://www.ifpennergiesnouvelles.com/article/biofuels-dashboard-2023> (accessed on 4 October 2024).
160. Watson, M.J.; Machado, P.G.; Da Silva, A.V.; Saltar, Y.; Ribeiro, C.O.; Nascimento, C.A.O.; Dowling, A.W. Sustainable Aviation Fuel Technologies, Costs, Emissions, Policies, and Markets: A Critical Review. *J. Clean. Prod.* **2024**, *449*, 141472. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.